

Operational Metric-Source Representation and Composite Ledger Closure under AASC

A Distributional Stress-Energy Theorem from Complete Metric-Response
Witnesses, Quotient Descent, and Finite Internal-Exchange Cancellation

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Module identity. AASC-GRQM-MA-SRC-1.0, v1.1.0-attachment-corrected.

Program status. Standalone Module A of the frozen repair program for *Complete Certificate Closure for a Constraint-Native Gravity-Quantum Realization under AASC*. This module is reviewed as an independent mathematical/software object before any capstone reintegration.

Abstract

This manuscript constructs the metric-facing source representation required by the constraint-native gravity–quantum realization program. It preserves the AASC dependency order: a determinate nondegenerate target first forces the kernel $K_D = (\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{Irr}_D)$; the realized physical interior and its deformation-response roles are fixed next; only then is a faithful metric projection licensed. No metric tensor, action, stress-energy tensor, conservation law, or source functional is permitted to authorize the upstream physical role that it represents.

The new type lock has two levels. The true base subtype

$$Q_{\text{resp}}^{\text{src}} \subseteq Q_{\text{resp}}$$

contains exactly those upstream response classes for which a complete upstream metric-incidence certificate exists. The structured source space $\hat{Q}_{\text{src}}$ then retains the response presentation, its attachment certificate, and its complete operational witness. A witness is not freely paired with a response: its graph must equal the metric-response relation already determined by the upstream protocol ledger. In particular, the zero functional attaches only when every certified metric-incidence protocol has zero response. The manuscript therefore neither assumes that every deformation response is source-bearing nor confuses a structured refinement with a set-theoretic subtype.

The principal representation theorem proves that every complete operational metric-response witness is represented by a unique symmetric tensor-valued distribution,

$$\mathcal{W}_x(h) = \frac{1}{2} \langle \mathbf{T}_x, h \rangle, \quad \mathbf{T}_x \in \mathcal{D}'(S^2 TM \otimes |\Lambda|),$$

and conversely. Support, covariance, regularity class, and quotient identity descend exactly. This gives a constructed, natural, quotient-faithful map

$$q_g : \hat{Q}_{\text{src}} \longrightarrow \text{Src}_g^{\text{dist}}.$$

Where an action or local source functional exists, the constructed distribution agrees with the Hilbert stress-energy tensor; action notation is therefore a compatible subclass, not a primitive premise.

The second main theorem proves the exact Ward–divergence equivalence. For every compactly supported vector field X ,

$$\mathcal{W}_x(\mathcal{L}_X g) = -\langle J_x, X \rangle \iff \nabla_\mu \mathbf{T}_{x\,\nu}^\mu = J_{x,\nu}$$

in the distributional sense. Closed source continuation is the case $J_x = 0$; open, boundary, anomalous, and effective cases remain explicitly exchange-declared.

The third main theorem constructs composite closure. A completed composite certificate contains a finite sector family, every cross term exactly once, a signed internal-exchange incidence matrix, external exchange currents, support and boundary data, and no-double-counting checks. The total response functional and total tensor are finite sums of the certified sector objects. Column-balance of the incidence matrix cancels every internal current, leaving exactly the declared external exchange. An accompanying software object validates the finite incidence, coverage, and residual-current conditions of a machine-readable certificate; it does not replace the analytic representation proof.

The reportable endpoint is attachment-safe source admission, source-representation existence, quotient descent, and closed-or-declared-exchange composite ledger closure for $\hat{Q}_{\text{src}}$. The manuscript does not prove universal inertial–gravitational proportionality, a nonrelativistic charge contraction, an upstream gravity–quantum continuation, a unique complete certificate orbit, weak equivalence, global PDE existence, or smoothness without a regularity certificate.

Read This First: Proof Spine

$$\text{TargetAdeq}_D \implies K_D = (\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{Irr}_D) \implies A_D^+ \implies I_D \implies Q_{\text{resp}}.$$

Only after that upstream sequence is fixed, the attachment theorem is applied:

$$\text{upstream metric-incidence protocols} + \text{attachment completeness} \implies Q_{\text{resp}}^{\text{src}} \subseteq Q_{\text{resp}}.$$

The structured witness quotient then carries the source map:

$$\hat{Q}_{\text{src}} = \{(p, \mathfrak{a}_p, \mathfrak{w}_p) : \text{SrcBear}_g(p, \mathfrak{a}_p, \mathfrak{w}_p)\} / \sim_{\text{src}} \xrightarrow{q_g} \text{Src}_g^{\text{dist}}.$$

Finally:

$$\text{Ward/exchange certificate} \implies \nabla_\mu \mathbf{T}^\mu_\nu = J_\nu,$$

and

$$\text{finite completed composite} + \text{balanced internal incidence} \implies \nabla_\mu \mathbf{T}^\mu_{\text{tot}\nu} = J_\nu^{\text{ext}}.$$

The source tensor is therefore a downstream representation of already admitted source-bearing response. It is not the source of response standing. A later capstone may import this module only after preserving that order.

Exact claim status

Item	Status in this manuscript
Kernel, RPI, mass/inertia, metric licensing	Imported theorem-use packages; not rederived and not authorized by source tensors.
Base source-bearing subtype $Q_{\text{resp}}^{\text{src}}$	Constructed by the independent attachment theorem from upstream metric-incidence protocols.
Structured source quotient $\hat{Q}_{\text{src}}$	Retains attachment certificate and complete witness; domain of q_g , not a synonym for the base subtype.
Tensor representation	New theorem proved here by distributional duality.
Quotient descent and naturality	New theorems proved here.
Ward/divergence equivalence	New theorem proved here.
Composite internal-exchange cancellation	New finite algebraic theorem proved here and independently checked by software.
Action/Hilbert tensor	Compatibility corollary for the action-presentable subclass.
Smooth source tensor	Conditional regularity upgrade; not automatic.
Universal mass proportionality, joint continuation, unique orbit	Explicitly outside this module.

Keywords

AASC; metric source; stress-energy tensor; distributional tensor; source ledger; deformation response; quotient descent; diffeomorphism covariance; Ward identity; composite source; internal exchange; finite incidence complex; closed continuation.

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Part I

Dependency Lock and Exact Module Target

1 Proof-Class Lock and Kernel Dependency

1.1 The exact question

The paper asks one narrow question:

Given an admitted upstream deformation-response presentation in a faithful metric projection, what independently proves that a complete operational metric-response witness belongs to that presentation; what mathematical object represents the attached first response; and when does a finite composite of attached responses form a closed or exchange-declared metric source ledger?

The question begins after the realized physical interior has been fixed. It does not ask which metric or tensor generates physical source standing. That reversal would violate the no-backflow discipline established by the kernel and the realized-physics arc [1, 2, 3].

Definition 1.1 (Target-adequate source-representation act). A source-representation act is target adequate when it fixes:

1. the same admitted upstream response class;
2. the metric-projection envelope in which first response is tested;
3. the complete family of admissible compactly supported metric probes;
4. the act-time status of the response;
5. lawful redescription and boundary behavior;
6. whether the source lineage is closed or exchange-declared.

Proposition 1.2 (Kernel dependency). *Every target-adequate source-representation act already instantiates the AASC kernel.*

Proof. The response must refer to the same upstream object across probes, so reference is active. Probe use must be licensed or unlicensed, so admissibility is active. A response must be reusable in the source ledger, so standing is active. Later selection cannot retroactively create the response of the earlier act, so irreversibility is active. None of these roles is supplied by the metric tensor or the source distribution. $\square$

1.2 Locked dependency order

The theorem-use order is

$$\begin{aligned} \text{TargetAdeq}_D &\Longrightarrow K_D \Longrightarrow A_D^+ \Longrightarrow I_D \\ &\Longrightarrow \text{deformation-response role} \Longrightarrow \text{metric projection} \Longrightarrow \text{source representation}. \end{aligned}$$

The realized physical interior already contains non-null persistence, localization, local domains, comparison, bounded transport, field carriers, particle persistence, gauge-compatible presentation, deformation response, vacuum baseline, temporal ordering, and causal dependence [2]. Mass and inertia are upstream deformation-response roles, not primitive scalars [3]. The Einstein projection identifies stress-energy as the metric-facing extraction of those already admitted roles, while leaving the existence and closure of that extraction as a separate burden [4]. This manuscript discharges that burden only after constructing the attachment relation, the true base source-bearing subtype, and the structured source-witness quotient defined below.

Theorem 1.3 (No metric-source backflow). *No tensor distribution, action, Hilbert variation, source equation, or covariant conservation identity constructed in this paper may authorize:*

1. *the AASC kernel;*
2. *the realized physical interior;*
3. *particle persistence;*
4. *the deformation-response role;*
5. *the faithful metric projection.*

It may only represent source standing already admitted upstream.

Proof. Every listed downstream object presupposes a fixed response target, metric-projection domain, lawful redescription, and continuation status. Using it to authorize those conditions makes a downstream representative into a boundary selector. The fixed-domain consequence layer classifies that move as anchor motion, hidden tensor load, representative privilege, or same-act repair. $\square$

1.3 Module code and frozen endpoint

This module is governed by *AASC – GRQM – MA – SRC – 1.0*. The endpoint is frozen at the following six burdens:

1. construct the exact base source-bearing subtype $Q_{\text{resp}}^{\text{src}}$ and the structured source-witness quotient $\hat{Q}_{\text{src}}$, with an independent attachment theorem preventing arbitrary witness lifts;
2. represent every complete metric-response witness by a unique symmetric tensor distribution;
3. prove support, naturality, regularity typing, and quotient descent;
4. prove the Ward–divergence equivalence;
5. construct finite composite source tensors with exact internal-exchange cancellation;
6. provide an independently executable finite certificate checker and at least one nontrivial inhabited analytic example.

A referee may show that the manuscript fails one of these burdens. A referee may not replace the module target with universal mass proportionality, full quantum gravity, global PDE existence, or a unique capstone assignment orbit.

2 Theorem-Use Inputs and Scope Guards

2.1 Realized physical interior

The paper imports the RPI theorem only at role level:

$$K_D \wedge \text{FixedAdmRealPers}_D \wedge \text{RoleSeq}_D \implies I_D \wedge \text{Coher}(I_D) \wedge \text{Closure}(I_D) \wedge \text{Minimal}(I_D).$$

No standard-physics object is used to prove this implication.

2.2 Mass and deformation response

For a localized persistence class Π , the imported role is

$$\mu_D(\Pi) : \Delta_\Pi \rightarrow C_D, \quad \iota_D(\Pi) = \mu_D(\Pi)|_{\Delta_\Pi^{\text{tr}}}.$$

The paper uses only the conclusion that physically mass-relevant content must represent quotient-stable standing-preserving deformation response. It does not assume a numerical mass value, stress tensor, Lagrangian, mass shell, or equivalence principle [3].

MGN1 supplies the numerical and quotient firewall: a coordinate, normalization, scheme, fit, or downstream embedding cannot acquire standing unless it changes an admitted witness class [5]. Here that result blocks the promotion of an arbitrary tensor or scalar into source content.

2.3 Faithful metric projection

The Einstein theorem supplies the fixed metric target:

$$\Pi_g(\text{Gravity}_D) = (M, g, \nabla)$$

with the minimal metric-compatible first-response grammar and a source side that must be symmetric, covariant, closed or exchange-declared, and extracted from upstream source roles [4]. The present paper constructs that source side. It does not rederive the Einstein tensor normal form.

2.4 Transport and covariant differentiation

Standing-preserving transport and covariant differentiation are imported only to type lawful comparison of source witnesses across the fixed metric projection [6, 7]. They do not generate the metric-response functional.

2.5 Scope guards

The module does not claim:

1. that every deformation-response class belongs to $Q_{\text{resp}}^{\text{src}}$;
2. that every source witness has a local action or smooth tensor density;
3. global or local existence for arbitrary coupled source equations;
4. universal inertial-gravitational proportionality;
5. an upstream GR/QM continuation;
6. uniqueness of all possible source models with the same integrated charges;
7. a quantum-gravity realization or weak-equivalence theorem.

3 The Source-Bearing Subtype and the Attachment Theorem

3.1 Response presentations before metric representation

Let RespPres_D denote the class of admitted upstream deformation-response presentations in the fixed realized physical interior. A presentation $p \in \text{RespPres}_D$ contains its upstream target, carrier, admitted deformation acts, complete response-protocol ledger, response outputs, lawful redescription class, support/continuation envelope, and failure status. It does not contain a metric source tensor or an arbitrary metric-response functional by definition.

Definition 3.1 (Complete upstream response protocol signature). For $p \in \text{RespPres}_D$, write

$$\Sigma_{\text{resp}}(p) = (\Delta_p^{\text{adm}}, \text{Prot}_p, \text{Inc}_p, \text{Out}_p, \ell_p^{\text{resp}}),$$

where Prot_p is the complete family of admitted upstream response protocols, Inc_p records which deformation acts each protocol interrogates, Out_p is the protocol-output map on its declared codomains, and ℓ_p^{resp} is the upstream witness, boundary, transport, failure, and redescription ledger. Completeness means that every standing-bearing distinction available to the fixed upstream response target is represented in this signature.

Let

$$Q_{\text{resp}} = \text{RespPres}_D / \sim_{\text{resp}}$$

be the final upstream response quotient. The relation $\sim_{\text{resp}}$ identifies exactly those presentations that no admitted upstream response witness separates. MGN1's standing firewall requires every additional distinction to expose a new admitted witness or else collapse to coordinate, skin, calibration, or selector status [5].

Remark 3.2 (Why the whole response quotient is not the source domain). A deformation response may be standing-bearing without possessing a metric-facing first-response branch in the declared projection envelope. The response may be typed only in another projection, may be source-neutral at first metric order, or may require a richer/effective metric carrier. The source-admission problem is therefore the

problem of proving which upstream response protocols actually carry metric-incidence data; it is not solved by attaching a functional to every $p \in \text{RespPres}_D$.

3.2 Metric-projection anchors and upstream metric incidence

Fix a smooth, paracompact, Hausdorff n -manifold M , a smooth Lorentzian metric g , the Levi-Civita connection ∇ , an orientation density bundle $|\Lambda|M$, and the admitted boundary class. Dimension, signature, differentiability, boundary status, and metric compatibility are projection anchors. They do not generate the upstream response role.

Definition 3.3 (Compact metric-variation space). The complete first-response probe space at g is

$$\mathcal{V}_g := \Gamma_c(S^2 T^* M),$$

the LF topological vector space of smooth compactly supported symmetric covariant two-tensors. Elements $h \in \mathcal{V}_g$ are infinitesimal covariant metric variations.

The use of all compactly supported symmetric variations is load-bearing. If the probe space were restricted to trace-free, divergence-free, homogeneous, or symmetry-adapted variations, distinct source tensors could have identical restricted response. Completeness of the probe family is what removes probe-invisible local source ambiguity.

Definition 3.4 (Faithful metric tangent map). For $p \in \text{RespPres}_D$, let $\mathcal{P}_p$ be its upstream response carrier and let $\Pi_g : \mathcal{P}_p \rightarrow \text{Met}(M)$ be the already licensed faithful metric projection on the declared metric branch. An admitted deformation act $\delta \in \Delta_p^{\text{adm}}$ belongs to the metric-tangent domain precisely when it is represented by an admitted C^1 response path $\gamma_\delta : (-\epsilon, \epsilon) \rightarrow \mathcal{P}_p$, with $\gamma_\delta(0) = p$, for which $\Pi_g \circ \gamma_\delta$ is differentiable at zero and its derivative is compactly supported. Define

$$\tau_{g,p}(\delta) := \left. \frac{d}{ds} \right|_{s=0} \Pi_g(\gamma_\delta(s)) \in \mathcal{V}_g.$$

If two admitted paths represent the same upstream deformation act, faithfulness of Π_g requires their derivatives to agree modulo metric-projection skin. The result is therefore a partial map

$$\tau_{g,p} : \Delta_p^{\text{adm}} \rightarrow \mathcal{V}_g$$

fixed by the upstream deformation act and the already licensed projection. It is not chosen by a prospective source witness. Failure of path representation, differentiability, compact support, or path-independence leaves the act outside the frozen first-response source envelope.

Proposition 3.5 (No independent tangent selector). *For fixed p and fixed faithful metric projection, the metric tangent of an admitted deformation act is unique up to metric-projection skin. A second map that assigns a different $h \in \mathcal{V}_g$ to the same upstream act either changes the projection tensor content or changes the projection envelope; it is not an alternative attachment certificate on the same target.*

Proof. The faithful projection preserves the upstream deformation identity and its first-order metric image. Two distinct tensor-level images are separated by a metric test variation and therefore cannot be projection skin. If the projection or its tangent domain is changed, an anchor or scope has moved. Hence the tangent map used by attachment is inherited rather than selected. $\square$

Definition 3.6 (Upstream metric-incidence relation). For $p \in \text{RespPres}_D$, define

$$\text{MetInc}_p(h, \pi) \iff \exists \delta \in \Delta_p^{\text{adm}} [\text{Inc}_p(\pi, \delta) \wedge \tau_{g,p}(\delta) = h].$$

Thus the relation between test variations and protocols is derived from the complete upstream protocol signature and the faithful metric tangent map. It is not an additional relation that may be freely attached downstream.

Only protocols whose output is a real first-response readout enter the present source envelope. Their independently determined metric-response relation is

$$\text{MetResp}_p := \{(h, a) \in \mathcal{V}_g \times \mathbb{R} : \exists \pi \in \text{Prot}_p, \text{MetInc}_p(h, \pi) \wedge \text{Out}_p(\pi) = a\}.$$

The relation may be partial, multivalued, discontinuous, or nonlinear. Those are attachment failures or richer-scope signals; no functional is supplied merely by writing the relation.

Definition 3.7 (Complete metric-attachment certificate). A complete metric-attachment certificate for p is

$$\mathbf{a}_p = (\tau_{g,p}, \text{Comp}_p, \text{Nat}_p^{\text{att}}, \text{Complete}_p^{\text{att}})$$

certifying the following properties of the already determined relation MetResp_p .

1. **Upstream provenance.** Every incident protocol and deformation act belongs to $\Sigma_{\text{resp}}(p)$, and every incidence is generated by $\tau_{g,p}$ through definition 3.6; no protocol, tangent, incidence, or output is added by the metric source construction.
2. **Totality.** For every $h \in \mathcal{V}_g$, at least one admitted protocol π satisfies $\text{MetInc}_p(h, \pi)$.
3. **Coherence.** If π and π' are both incident to the same h , then $\text{Out}_p(\pi) = \text{Out}_p(\pi')$.
4. **First-order vector compatibility.** The certified protocol composition and rescaling operations carry (h_1, h_2) and (λ, h) to protocols incident to $h_1 + h_2$ and λh , with outputs equal to the corresponding sum and scalar multiple.
5. **LF continuity.** The single-valued response determined by totality and coherence is continuous in the LF topology on $\mathcal{V}_g$.
6. **Lawful transport.** Lawful response and metric redescription transport the incidence relation and output relation naturally.
7. **Metric-protocol completeness.** Every upstream protocol in the fixed response signature whose faithful metric tangent lies in $\mathcal{V}_g$ occurs in the incidence relation. No metric-facing response channel is omitted.

Theorem 3.8 (Operational source-attachment theorem). *Let $p \in \text{RespPres}_D$ carry a complete metric-attachment certificate $\mathbf{a}_p$. Then MetResp_p is the graph of a unique continuous linear functional*

$$\mathcal{W}_p : \mathcal{V}_g \longrightarrow \mathbb{R}.$$

Conversely, a continuous linear functional is attached to p by $\mathbf{a}_p$ only when its graph is exactly MetResp_p .

Proof. Totality and coherence make MetResp_p the graph of a unique function. First-order vector compatibility makes that function additive and homogeneous. The continuity clause makes it a member of $\mathcal{L}_{\text{cont}}(\mathcal{V}_g, \mathbb{R})$. Uniqueness follows because a function is fixed by its graph. The converse is the attachment requirement: a functional whose graph differs at even one test variation does not reproduce the complete upstream metric-response relation and is not a witness of p . $\square$

Definition 3.9 (Source-attachment relation). Write

$$\text{Attach}_g(p, \mathbf{a}_p, \mathcal{W})$$

iff $\mathbf{a}_p$ is a complete metric-attachment certificate for p and

$$\text{Graph}(\mathcal{W}) = \text{MetResp}_p.$$

The relation is prior to distributional source representation. It determines which operational functional belongs to which upstream response presentation.

Corollary 3.10 (Certified-zero criterion). *For a complete attachment certificate $\mathbf{a}_p$,*

$$\text{Attach}_g(p, \mathbf{a}_p, 0)$$

holds iff every protocol incident to every $h \in \mathcal{V}_g$ has upstream output zero. Thus a zero source witness is admissible only as the independently certified actual first metric response of p , not as an arbitrary functional attached by notation.

Proof. The zero functional has graph $\mathcal{V}_g \times \{0\}$. Apply definition 3.9 and the definition of MetResp_p . $\square$

Corollary 3.11 (No metric protocol, no attachment). *If some $h \in \mathcal{V}_g$ has no incident upstream protocol, or if two incident protocols at one h have distinct outputs, then no source attachment exists in the frozen Module A envelope.*

Proposition 3.12 (Response-equivalence invariance). *If $p \sim_{\text{resp}} p'$, the complete upstream protocol signatures are transported lawfully, and $\text{Attach}_g(p, \mathbf{a}_p, \mathcal{W}_p)$, then there are transported data $\mathbf{a}_{p'}$ and $\mathcal{W}_{p'}$ satisfying $\text{Attach}_g(p', \mathbf{a}_{p'}, \mathcal{W}_{p'})$. Hence attachability is a property of the upstream response class, not of a representative label.*

Proof. Response equivalence preserves every admitted upstream protocol output and lawful transport preserves the metric-incidence relation. Therefore the transported metric-response relation is the transported graph of $\mathcal{W}_p$, and theorem 3.8 supplies the unique transported functional. $\square$

3.3 The true base subtype and the structured source quotient

Definition 3.13 (Complete operational metric witness). Given $\text{Attach}_g(p, \mathbf{a}_p, \mathcal{W}_p)$, a complete operational metric witness is

$$\mathfrak{w}_p = (\mathcal{W}_p, K_p, \text{Nat}_p, \text{Ward}_p, \text{Reg}_p, \partial_p)$$

with the following additional certificates.

1. $K_p \subseteq M$ is closed and $\mathcal{W}_p(h) = 0$ whenever $\text{supp } h \cap K_p = \emptyset$.
2. Nat_p records lawful-redescription naturality.

3. Ward_p is a continuous vector-distribution J_p satisfying

$$\mathcal{W}_p(\mathcal{L}_X g) = -\langle J_p, X \rangle$$

for every compactly supported smooth vector field X .

4. Reg_p records the exact distributional, measure, L^1_{loc} , Sobolev, or smooth regularity actually carried.
5. ∂_p records whether compact support avoids the boundary or a declared boundary response is included.

Definition 3.14 (Source-bearing predicate and presentation). A source-bearing presentation is a triple

$$(p, \mathfrak{a}_p, \mathfrak{w}_p)$$

for which

$$\text{SrcBear}_g(p, \mathfrak{a}_p, \mathfrak{w}_p) \iff \text{Attach}_g(p, \mathfrak{a}_p, \mathcal{W}_p) \wedge \text{CompleteMetricWitness}_g(\mathfrak{w}_p).$$

The class of all such triples is SrcPres_D .

Definition 3.15 (Base source-bearing subtype). The genuine source-bearing subtype of the upstream response quotient is

$$Q_{\text{resp}}^{\text{src}} := \{[p]_{\sim_{\text{resp}}} \in Q_{\text{resp}} : \exists \mathfrak{a}_p, \mathfrak{w}_p \text{ SrcBear}_g(p, \mathfrak{a}_p, \mathfrak{w}_p)\} \subseteq Q_{\text{resp}}.$$

Well-definedness follows from proposition 3.12 and lawful transport of the witness ledgers.

Definition 3.16 (Source-response equivalence). Two source-bearing presentations are source-equivalent,

$$(p, \mathfrak{a}_p, \mathfrak{w}_p) \sim_{\text{src}} (p', \mathfrak{a}_{p'}, \mathfrak{w}_{p'}),$$

iff:

1. $p \sim_{\text{resp}} p'$;
2. after lawful transport, their upstream metric-response relations coincide;
3. their attached functionals agree on all of $\mathcal{V}_g$;
4. their Ward/exchange distributions, support, regularity, and boundary ledgers agree up to lawful equivalence.

Protocol names and representative incidence labels may differ when they carry the same transported upstream response relation.

Definition 3.17 (Structured source-witness quotient). The exact domain of the metric-source representation map is the structured quotient

$$\widehat{Q}_{\text{src}} := \text{SrcPres}_D / \sim_{\text{src}}.$$

It is a structured refinement over, not a set-theoretic synonym for, the base subtype $Q_{\text{resp}}^{\text{src}}$. The forgetful map

$$F : \widehat{Q}_{\text{src}} \longrightarrow Q_{\text{resp}}^{\text{src}}, \quad [(p, \mathfrak{a}_p, \mathfrak{w}_p)]_{\sim_{\text{src}}} \longmapsto [p]_{\sim_{\text{resp}}},$$

is well defined and surjective. Distinct attached source structures may lie above one base class until a later fibre-uniqueness theorem identifies them.

Proposition 3.18 (No hidden source promotion). *A response class does not enter $Q_{\text{resp}}^{\text{src}}$ merely because a scalar, energy label, mass value, Hamiltonian coefficient, fitted tensor, or zero functional is written beside it. Entry requires a complete upstream attachment certificate and a complete operational metric witness.*

Proof. The metric functional must equal the graph already determined by the upstream metric-incidence protocols. A fitted or arbitrarily selected functional fails definition 3.9. A genuinely new response requires a new admitted upstream protocol and therefore changes the response signature rather than silently promoting the old class. $\square$

Proposition 3.19 (The subtype condition is non-tautological). *There are well-typed response presentations for which the attachment predicate fails. In particular, a response presentation whose complete upstream protocol ledger contains no metric-incidence protocol has a class in Q_{resp} but no class in $Q_{\text{resp}}^{\text{src}}$.*

Proof. Such a presentation may carry a standing-bearing response under a nonmetric protocol while MetInc_p is empty. Totality in definition 3.7 fails, so corollary 3.11 gives no attached functional. A finite explicit model is given in example 13.1. $\square$

3.4 Closed, open, and anomalous status

Definition 3.20 (Closed, open, and anomalous source status). For $x = [(p, \mathfrak{a}_p, \mathfrak{w}_p)] \in \widehat{Q}_{\text{src}}$:

1. x is *closed* when $J_p = 0$;
2. x is *exchange-declared* when $J_p \neq 0$ and J_p is assigned to a named external, boundary, open, or effective ledger;
3. x is *anomalous or incomplete* when the Ward residual is nonzero but no declared ledger carries it.

This classification is made after attachment and before conservation is claimed. It prevents the words “source-bearing” and “closed” from functioning as substitutes for, respectively, an attachment proof and a divergence proof.

Part II

Operational Source Representation

4 Complete Metric Probes and Operational Witnesses

4.1 Topology of the probe space

For every compact $K \subset M$, let

$$\mathcal{V}_{g,K} := \{h \in \Gamma^\infty(S^2T^*M) : \text{supp } h \subseteq K\}$$

with its standard Fréchet topology generated by the C^r -seminorms on K . Then

$$\mathcal{V}_g = \varinjlim_K \mathcal{V}_{g,K}$$

with the usual LF topology. A functional $\mathcal{W} : \mathcal{V}_g \rightarrow \mathbb{R}$ is continuous precisely when each restriction to $\mathcal{V}_{g,K}$ is continuous with a finite-order seminorm bound on every compact K [8, 9].

Lemma 4.1 (Finite-order local bound). *For every continuous $\mathcal{W} : \mathcal{V}_g \rightarrow \mathbb{R}$ and compact $K \subset M$, there exist an integer r , a constant C_K , and a finite atlas over K such that*

$$|\mathcal{W}(h)| \leq C_K \sum_{j=0}^r \sup_K |\nabla^j h| \quad (h \in \mathcal{V}_{g,K}).$$

Proof. This is the standard local continuity characterization of distributions on the compactly supported test-section space. The covariant-derivative seminorms are equivalent on a fixed compact set to coordinate derivative seminorms, so the statement is independent of the chosen finite atlas. $\square$

Definition 4.2 (Complete metric probe). A probe family $\mathcal{P} \subseteq \mathcal{V}_g$ is complete when its linear span is dense in $\mathcal{V}_g$ and every continuous linear functional vanishing on $\mathcal{P}$ vanishes on all of $\mathcal{V}_g$.

The canonical module uses $\mathcal{P} = \mathcal{V}_g$. Smaller experimentally or symmetry selected families may be useful, but they do not carry the uniqueness theorem proved below.

Lemma 4.3 (Probe-completeness necessity). *If $\mathcal{P}$ is not complete, there exist distinct tensor distributions $\mathbf{T} \neq \mathbf{T}'$ whose pairings agree on every $h \in \mathcal{P}$.*

Proof. Non-completeness means that the closure of the span of $\mathcal{P}$ is a proper closed subspace of $\mathcal{V}_g$. By Hahn–Banach there is a nonzero continuous linear functional L annihilating that subspace. Given any distribution $\mathbf{T}$, let $\mathbf{T}'$ represent the functional $h \mapsto \frac{1}{2}\langle \mathbf{T}, h \rangle + L(h)$. Then $\mathbf{T}' \neq \mathbf{T}$ but the two pairings agree on $\mathcal{P}$. $\square$

4.2 Operational origin of the attached witness

The source functional is not free input. It must be the unique function whose graph is the upstream metric-response relation constructed in definition 3.6 and theorem 3.8. Two useful realization routes are available, but both must discharge that same attachment burden.

Definition 4.4 (Attached differentiable metric-facing response carrier). An attached metric-facing response carrier for p is a scalar-valued map

$$\mathcal{R}_p : \mathcal{U}_g \longrightarrow \mathbb{R}$$

on an open neighborhood $\mathcal{U}_g$ of g in the fixed-signature metric space, together with a complete attachment certificate $\mathbf{a}_p$, such that:

1. the finite metric-deformation readouts of $\mathcal{R}_p$ are protocols in $\Sigma_{\text{resp}}(p)$ and generate the certified relation MetResp_p ;
2. $\mathcal{R}_p$ is Fréchet or Hadamard differentiable at g on every compact-support Fréchet step, with compatible derivatives; alternatively, if a Gâteaux directional derivative is used, the certificate separately proves that it is additive, homogeneous, and LF-continuous;

3. lawful metric redescription preserves both $\mathcal{R}_p$ and the upstream incidence relation;
4. the resulting derivative graph equals MetResp_p .

Proposition 4.5 (Attached operational linearization). *Every attached differentiable metric-facing response carrier induces the unique functional*

$$\mathcal{W}_p(h) := \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \mathcal{R}_p(g + \epsilon h)$$

satisfying $\text{Attach}_g(p, \mathbf{a}_p, \mathcal{W}_p)$.

Proof. The differentiability certificate yields a continuous linear derivative on $\mathcal{V}_g$. The final attachment clause identifies its graph with the complete upstream metric-response relation. The uniqueness statement is therefore theorem 3.8, not a generic property of an arbitrarily chosen derivative. $\square$

Definition 4.6 (Direct attached first-response ledger). A direct first-response ledger is admitted only when it supplies a complete attachment certificate $\mathbf{a}_p$ and a continuous linear functional $\mathcal{W}_p$ satisfying

$$\text{Graph}(\mathcal{W}_p) = \text{MetResp}_p.$$

Continuity, linearity, support, naturality, Ward, regularity, and boundary data do not by themselves attach the ledger to p ; graph equality with the upstream protocol response is indispensable.

Remark 4.7 (Two admission routes, one attachment theorem). A source presentation may expose a differentiable carrier whose derivative realizes the upstream relation, or may expose the operational ledger directly. In both routes, Attach_g is proved before distributional duality. The zero ledger is admitted only under the certified-zero criterion of corollary 3.10.

4.3 Support and locality status

Definition 4.8 (Witness support). The support of a witness is the complement of the largest open set $U \subseteq M$ such that $\mathcal{W}(h) = 0$ for every h supported in U .

Proposition 4.9 (Support is not a primitive point set). *The support of $\mathcal{W}_p$ is fixed by the response of the complete metric probe family. Choosing a smaller region without changing any probe response is skin; choosing a smaller region that deletes a nonzero response is source-data loss.*

Proof. The support is an extensional property of the functional. It is therefore invariant under presentation changes and cannot be altered by an independent locator without changing the functional or merely relabeling it. $\square$

4.4 No hidden action assumption

The operational witness is deliberately more general than an action derivative. Dissipative effective descriptions, distributional bodies, algebraic expectation values, and response kernels may possess a complete first metric response without being given by one primitive local matter action. When an action exists, it becomes a compatible representation route proved in section 7.

5 Distributional Source Representation

5.1 Tensor distributions

Let

$$E := S^2 T^* M.$$

The continuous dual of $\Gamma_c(E)$ is canonically the space of distributional sections of

$$E^* \otimes |\Lambda| M \cong S^2 TM \otimes |\Lambda| M.$$

We write this space as $\mathcal{D}'(S^2 TM \otimes |\Lambda|)$ [8, 9].

Theorem 5.1 (Operational Metric-Source Representation). *For every continuous linear operational witness*

$$\mathcal{W} : \mathcal{V}_g \rightarrow \mathbb{R}$$

there exists a unique symmetric contravariant tensor density distribution

$$\mathbf{T}_{\mathcal{W}} \in \mathcal{D}'(S^2 TM \otimes |\Lambda|)$$

such that

$$\mathcal{W}(h) = \frac{1}{2} \langle \mathbf{T}_{\mathcal{W}}, h \rangle \quad \text{for every } h \in \mathcal{V}_g.$$

Conversely, every $\mathbf{T} \in \mathcal{D}'(S^2 TM \otimes |\Lambda|)$ defines a unique continuous linear witness by this formula.

Proof. By the definition of the distribution space of sections, every continuous linear functional on $\Gamma_c(E)$ is a unique element of $\mathcal{D}'(E^* \otimes |\Lambda| M)$. Apply the canonical bundle isomorphism $E^* \cong S^2 TM$ and multiply the representing distribution by 2 to obtain the stated normalization. Conversely, pairing any tensor density distribution with a compactly supported test section gives a continuous linear functional. Uniqueness is equality in the continuous dual. $\square$

Corollary 5.2 (No tensor selector). *A complete operational witness admits exactly one local distributional source tensor. No coordinate, basis, improvement convention, or representative choice can select a different tensor while preserving all compact metric responses.*

Proof. Any different tensor distribution would define a different continuous functional on $\mathcal{V}_g$ by theorem 5.1. If all probe responses are unchanged, the difference distribution pairs to zero with every test section and is therefore zero. $\square$

Definition 5.3 (Lowered source distribution). The smooth metric g lowers the indices of $\mathbf{T}_{\mathcal{W}}$ distributionally. We denote the resulting covariant tensor density by

$$\mathbf{T}_{\mathcal{W}, \mu\nu} = g_{\mu\alpha} g_{\nu\beta} \mathbf{T}_{\mathcal{W}}^{\alpha\beta}.$$

When the density factor is separated using dvol_g , the notation $T_{\mu\nu}$ is used for the corresponding tensor distribution.

5.2 Support and order

Theorem 5.4 (Support identity).

$$\text{supp } \mathbf{T}_{\mathcal{W}} = \text{supp } \mathcal{W}.$$

In particular, a witness localized to a closed set K produces a source distribution supported in K , and no smaller support is compatible with the same complete witness.

Proof. Both supports are defined as the complement of the largest open set on which the pairing with every supported test section vanishes. The representation identity makes those open sets identical. $\square$

Proposition 5.5 (Distribution order). *The local order of $\mathbf{T}_{\mathcal{W}}$ on a compact K is bounded by the least integer r for which the estimate in lemma 4.1 holds.*

Proof. This is the standard order definition for a distribution in local coordinates, transported through the tensor bundle trivialization. $\square$

5.3 Representation equivalence

Theorem 5.6 (Operational source representation equivalence). *The assignments*

$$\mathcal{W} \mapsto \mathbf{T}_{\mathcal{W}}, \quad \mathbf{T} \mapsto (h \mapsto \tfrac{1}{2}\langle \mathbf{T}, h \rangle)$$

are inverse linear isomorphisms between the space of continuous operational metric first-response witnesses and $\mathcal{D}'(S^2TM \otimes |\Lambda|)$.

Proof. Both compositions are the identity by theorem 5.1. $\square$

Remark 5.7 (What has and has not been constructed). The theorem constructs the metric source tensor from operational response. It does not construct an operational witness for a response class that lacks one. That is why the domain is $\hat{Q}_{\text{src}}$, not Q_{resp} .

5.4 Regularity stratification

Definition 5.8 (Regular source classes). Let:

$$\begin{aligned} \text{Src}_g^{\text{dist}} &:= \mathcal{D}'(S^2TM \otimes |\Lambda|), \\ \text{Src}_g^{\text{meas}} &:= \{\mathbf{T} \in \text{Src}_g^{\text{dist}} : \mathbf{T} \text{ is a tensor-valued Radon measure}\}, \\ \text{Src}_g^{L^1} &:= \{\mathbf{T} = T \, \text{dvol}_g : T \in L^1_{\text{loc}}(S^2TM)\}, \\ \text{Src}_g^{\text{sm}} &:= \{\mathbf{T} = T \, \text{dvol}_g : T \in C^\infty(S^2TM)\}. \end{aligned}$$

Proposition 5.9 (No automatic smoothness). *Continuity of $\mathcal{W}$ implies a tensor distribution and no stronger regularity. Measure, L^1_{loc} , Sobolev, or smooth status requires an explicit regularity certificate.*

Proof. Delta-supported and derivative-of-delta distributions are continuous on the test space but are not represented by smooth tensor fields. Therefore continuity alone cannot imply smoothness. $\square$

6 Quotient Descent, Naturality, and Faithfulness

6.1 Lawful metric redescription

Let

$$\phi : (M, g, p) \longrightarrow (M', g', p')$$

be an admitted orientation-preserving diffeomorphism satisfying $\phi_*g = g'$ and preserving the upstream response role. It acts on metric variations by pushforward and on tensor density distributions by duality:

$$\langle \phi_*\mathbf{T}, h' \rangle := \langle \mathbf{T}, \phi^*h' \rangle.$$

Theorem 6.1 (Naturality of the represented source). *If the operational witnesses are natural,*

$$\mathcal{W}_{p'}(\phi_*h) = \mathcal{W}_p(h),$$

then

$$\mathbf{T}_{p'} = \phi_*\mathbf{T}_p.$$

Proof. For every $h' \in \Gamma_c(S^2T^*M')$,

$$\frac{1}{2}\langle \mathbf{T}_{p'}, h' \rangle = \mathcal{W}_{p'}(h') = \mathcal{W}_p(\phi^*h') = \frac{1}{2}\langle \mathbf{T}_p, \phi^*h' \rangle = \frac{1}{2}\langle \phi_*\mathbf{T}_p, h' \rangle.$$

Uniqueness in theorem 5.1 gives the result. $\square$

Corollary 6.2 (Coordinate and gauge skin). *Coordinate changes and lawful representative changes cannot create, remove, or modify the represented source distribution. If a purported change alters $\mathbf{T}$, it altered standing-bearing metric response or moved the projection scope.*

6.2 The constructed quotient map

Definition 6.3 (Metric-source quotient map). Define

$$q_g : \widehat{Q}_{\text{src}} \longrightarrow \text{Src}_g^{\text{dist}} / \simeq_{\text{law}}$$

by

$$q_g([(p, \mathbf{a}_p, \mathbf{w}_p)]_{\sim_{\text{src}}}) := [\mathbf{T}_{\mathcal{W}_p}]_{\simeq_{\text{law}}},$$

where $\simeq_{\text{law}}$ is lawful metric redescription together with the declared boundary equivalence of the fixed projection envelope.

Theorem 6.4 (Well-defined quotient descent). *The map q_g is well defined.*

Proof. Let two representatives be source-equivalent. After lawful transport to one metric envelope their witnesses agree on all of $\mathcal{V}_g$. By uniqueness in theorem 5.1, their tensor distributions agree. Naturality theorem 6.1 then transports that equality between lawful metric presentations. Their support, Ward, regularity, and boundary statuses also agree by definition 3.16. Hence their target classes are the same. $\square$

Theorem 6.5 (Probe faithfulness). *For $x, y \in \widehat{Q}_{\text{src}}$ represented in one metric envelope,*

$$q_g(x) = q_g(y) \iff \mathcal{W}_x(h) = \mathcal{W}_y(h) \text{ for every } h \in \mathcal{V}_g,$$

modulo the fixed lawful-redescription and boundary equivalence.

Proof. Equality of tensor distributions implies equality of all pairings. Equality of all pairings implies equality of the distributions by theorem 5.1. The quotient qualification is handled by theorem 6.1. $\square$

Corollary 6.6 (No source ambiguity below the complete probe quotient). *Within $\widehat{Q}_{\text{src}}$, two presentations over the same base class in $Q_{\text{resp}}^{\text{src}}$ but with inequivalent attachment-certified metric-response relations are distinct source-bearing classes. Two presentations with the same transported attachment relation and complete witness have the same metric source tensor.*

This corollary makes the scope precise. The paper does not claim that the upstream response quotient alone determines source response. It claims that the exact source-bearing refinement, whose standing difference is operationally exposed by metric probes, has a unique standard tensor representation.

6.3 Improvement and boundary ambiguities

Definition 6.7 (Interior-null modification). A tensor distribution $\mathbf{I}$ is interior-null when

$$\langle \mathbf{I}, h \rangle = 0$$

for every compactly supported metric variation h in the admitted interior probe space.

Proposition 6.8 (Interior-null modifications vanish). *Every interior-null tensor distribution is zero on the interior. A nonzero “improvement” can survive only by:*

1. *changing the complete interior metric response;*
2. *living on a declared boundary;*
3. *moving to a restricted probe family;*
4. *changing the source or projection scope.*

Proof. Vanishing on all compactly supported interior test sections is the definition of the zero distribution on the interior. Boundary-supported distributions are invisible only when the test space is chosen to avoid the boundary, in which case their status must be recorded separately. $\square$

Definition 6.9 (Boundary source component). On a manifold with boundary, the complete source object is

$$(\mathbf{T}_{\text{bulk}}, \mathbf{T}_{\partial}, \mathcal{B}),$$

where $\mathbf{T}_{\partial}$ is the boundary distribution and $\mathcal{B}$ fixes the permitted boundary test variations. Omitting a nonzero boundary response is ledger incompleteness, not an improvement convention.

6.4 Image and exact interface

Definition 6.10 (Represented source image).

$$\text{Im}(q_g) = \{ [\mathbf{T}_{\mathcal{W}}]_{\simeq_{\text{law}}} : [(p, \mathbf{a}_p, \mathbf{w}_p)] \in \widehat{Q}_{\text{src}} \}.$$

Theorem 6.11 (Exact source-image characterization). *A lawful tensor-distribution class belongs to $\text{Im}(q_g)$ iff there exists an attachment-certified structured source presentation $(p, \mathbf{a}_p, \mathbf{w}_p)$ whose complete witness is the tensor pairing and whose support, naturality, Ward/exchange, regularity, and boundary coordinates satisfy the source-admission interface.*

Proof. The forward implication follows from the construction. For the converse, the hypothesis already supplies $\text{Attach}_g(p, \mathfrak{a}_p, \mathcal{W}_p)$; the tensor pairing therefore reproduces the independently fixed upstream metric-response graph rather than creating an attachment. The representation equivalence theorem 5.6 returns the same tensor class. $\square$

7 Regularity Upgrades and Action Compatibility

7.1 Regularity certificates

Definition 7.1 (Density representation certificate). A witness has a C^k density certificate when there exists

$$T^{\mu\nu} \in C_{\text{loc}}^k(S^2TM)$$

such that

$$\mathcal{W}(h) = \frac{1}{2} \int_M T^{\mu\nu} h_{\mu\nu} \, \text{dvol}_g$$

for every $h \in \mathcal{V}_g$.

Theorem 7.2 (Regularity upgrade). *If $\mathcal{W}$ has a C^k density certificate, its representing distribution is*

$$\mathbf{T}_{\mathcal{W}} = T^{\mu\nu} \text{dvol}_g$$

and has that regularity. Conversely, if the representing distribution is induced by a C^k tensor density, the displayed formula is the unique C^k density certificate.

Proof. The formula defines the same continuous functional as $\mathbf{T}_{\mathcal{W}}$; uniqueness in theorem 5.1 identifies them. The converse is immediate from the distributional pairing. $\square$

Corollary 7.3 (Smooth Einstein-interface source). *The classical smooth Einstein-source interface is the subdomain*

$$\widehat{Q}_{\text{src}}^{\text{sm}} := \{x \in \widehat{Q}_{\text{src}} : q_g(x) \in \text{Src}_g^{\text{sm}}\}.$$

The module does not promote a distributional source to this interface without a smooth regularity certificate.

7.2 Action-presentable subclass

Definition 7.4 (Action-presentable source). A source-bearing presentation is action-presentable when there is a functional

$$\mathcal{S}_p[g, \Psi]$$

defined on a declared field and metric domain such that:

1. the metric derivative exists and is continuous for compactly supported variations;
- 2.

$$\mathcal{W}_p(h) = \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \mathcal{S}_p[g + \epsilon h, \Psi];$$

3. the functional is lawfully diffeomorphism covariant;
4. all boundary and field-domain data are declared.

Theorem 7.5 (Hilbert compatibility). *For an action-presentable source,*

$$\mathbf{T}_{\mathcal{W}} = 2 \frac{\delta \mathcal{S}_p}{\delta g}$$

as a tensor density distribution, with the sign and index convention fixed by definition 7.4. In the conventional inverse-metric convention this is equivalently

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta \mathcal{S}_p}{\delta g^{\mu\nu}}.$$

Proof. The variational derivative is, by definition, the distribution representing the continuous first variation. It therefore represents the same witness as $\mathbf{T}_{\mathcal{W}}$. Uniqueness gives equality. $\square$

Remark 7.6 (Action notation is downstream). The theorem does not infer source standing from the action. The source-bearing response and complete operational witness are already fixed. The action is one presentation that generates the same witness.

7.3 Local functionals

Suppose

$$\mathcal{S}_p[g, \Psi] = \int_M L(g, \Psi, \nabla \Psi, \dots, \nabla^r \Psi) \, \text{dvol}_g$$

with declared compact support or boundary conditions. Standard variational calculus then gives a finite-order distribution and, under smooth field/domain hypotheses, a smooth Hilbert tensor [10, 11]. The module uses this only as a regularity route, not as the definition of source-bearing response.

Proposition 7.7 (A concrete action does not prove class-wide existence). *One action-presentable example proves that $\hat{Q}_{\text{src}}$ is nonempty. It does not prove that every element of Q_{resp} or every source-bearing class is action-presentable.*

Proof. Existence of one representative in a subclass is not a surjectivity theorem onto the whole source-bearing quotient. $\square$

7.4 Distributional bodies

Worldline, shell, defect, and sharp-interface sources may be naturally distributional. They are legitimate members of $\hat{Q}_{\text{src}}$ when their complete metric-response witness is continuous and their Ward/exchange status is certified. They enter the smooth Einstein interface only after an appropriate regularization

or weak-solution theorem; the present module does not erase that distinction [12].

Part III

Ward Identity and Continuation Status

8 Ward Profiles and Distributional Divergence

8.1 Covariant divergence of a tensor distribution

The smooth Levi-Civita connection extends to tensor distributions by duality. For a symmetric contravariant tensor density distribution $\mathbf{T}^{\mu\nu}$, define its divergence by

$$\langle \nabla_\mu \mathbf{T}^\mu{}_\nu, X^\nu \rangle := -\langle \mathbf{T}^\mu{}_\nu, \nabla_\mu X^\nu \rangle$$

for every compactly supported smooth vector field X .

Lemma 8.1 (Lie-derivative pairing). *For every symmetric tensor distribution $\mathbf{T}$ and compactly supported smooth vector field X ,*

$$\frac{1}{2} \langle \mathbf{T}, \mathcal{L}_X g \rangle = -\langle \nabla_\mu \mathbf{T}^\mu{}_\nu, X^\nu \rangle.$$

Proof. Metric compatibility gives

$$(\mathcal{L}_X g)_{\mu\nu} = \nabla_\mu X_\nu + \nabla_\nu X_\mu.$$

Symmetry of $\mathbf{T}^{\mu\nu}$ reduces half the pairing to $\langle \mathbf{T}^{\mu\nu}, \nabla_\mu X_\nu \rangle$. The definition of the distributional covariant derivative gives the stated negative divergence pairing. Compact support removes boundary terms. $\square$

Theorem 8.2 (Ward–Divergence Equivalence). *Let $x \in \widehat{Q}_{\text{src}}$ have operational witness $\mathcal{W}_x$, represented source $\mathbf{T}_x$, and declared exchange current distribution J_x . Then*

$$\mathcal{W}_x(\mathcal{L}_X g) = -\langle J_x, X \rangle \quad \text{for all } X \in \Gamma_c(TM)$$

iff

$$\nabla_\mu \mathbf{T}_x^\mu{}_\nu = J_{x,\nu}$$

as vector-valued distributions.

Proof. By theorem 5.1 and lemma 8.1,

$$\mathcal{W}_x(\mathcal{L}_X g) = \frac{1}{2} \langle \mathbf{T}_x, \mathcal{L}_X g \rangle = -\langle \nabla_\mu \mathbf{T}_x^\mu{}_\nu, X^\nu \rangle.$$

Equality with $-\langle J_x, X \rangle$ for every compactly supported vector field is equivalent to equality of the two vector distributions. $\square$

Corollary 8.3 (Closed source continuation). *The source class is closed precisely when*

$$\mathcal{W}_x(\mathcal{L}_X g) = 0 \quad \text{for every compactly supported } X,$$

equivalently

$$\nabla_\mu \mathbf{T}^\mu_{x\nu} = 0.$$

Corollary 8.4 (Declared exchange). *For an open or effective source,*

$$\nabla_\mu \mathbf{T}^\mu_{x\nu} = J^\text{declared}_\nu$$

is not a failure of source representation. It is the exact metric form of the declared exchange coordinate in the source witness.

8.2 Noether compatibility without action primacy

If the witness comes from a diffeomorphism-covariant action, its infinitesimal diffeomorphism variation yields the same Ward profile, with field-equation and boundary terms exposed in the usual way [10, 13]. The operational theorem above is logically prior: it needs only the complete metric first-response witness and its Ward coordinate.

Theorem 8.5 (Action-based Ward compatibility). *Let an action-presentable source have fields Ψ^A , Euler–Lagrange distributions E_A , boundary/external current J^ext_ν , and diffeomorphism-covariant action. Then its represented tensor satisfies an off-shell identity of the form*

$$\nabla_\mu T^\mu{}_\nu = \sum_A E_A \mathcal{D}_\nu \Psi^A + J^\text{ext}_\nu,$$

where $\mathcal{D}_\nu \Psi^A$ is the infinitesimal diffeomorphism action on the field. On a certified solution, this reduces to the Ward profile of theorem 8.2.

Proof. Differentiate the diffeomorphism-covariance identity for the action along a compactly supported vector field. Decompose the variation into the metric first variation, field variations, and declared boundary/external terms. Use theorem 7.5 and lemma 8.1 and arbitrariness of X . This is the standard covariant Noether identity, here used only in the action-presentable subclass. $\square$

Remark 8.6 (What on-shell means here). An “on-shell” claim is licensed only when the source certificate supplies an actual field configuration or solution class on which all declared E_A vanish and the boundary/exchange terms have the stated status. Writing formal equations without a solution certificate does not establish closed source continuation.

8.3 Anomaly and incompleteness

Definition 8.7 (Ward residual). For a proposed source tensor and declared exchange J , define

$$\mathcal{A}_\nu := \nabla_\mu \mathbf{T}^\mu{}_\nu - J_\nu.$$

Proposition 8.8 (Residual classification). *If $\mathcal{A} \neq 0$, then at least one of the following holds:*

1. *the operational witness is not diffeomorphism compatible;*

2. *an internal or external exchange channel is omitted;*
3. *a boundary contribution is omitted;*
4. *an anomaly is present;*
5. *the proposed field configuration is off shell;*
6. *the source or metric scope changed;*
7. *standing fails.*

Proof. The Ward–divergence theorem is exact. A nonzero residual means the claimed witness profile and the represented tensor divergence disagree. The listed routes are the exhaustive locations at which the claimed profile can fail or be incomplete inside the fixed source interface. $\square$

9 Boundary, Open-System, and Exchange Discipline

9.1 Interior probes

On a manifold with boundary, the interior theorem uses metric variations and vector fields compactly supported away from ∂M . The representation and Ward theorems then hold on the interior without boundary terms.

Definition 9.1 (Boundary-complete source witness). A boundary-complete source witness consists of:

$$(\mathcal{W}_{\text{bulk}}, \mathcal{W}_{\partial}, \mathbf{T}_{\text{bulk}}, \mathbf{T}_{\partial}, J_{\text{bulk}}, J_{\partial}, \mathcal{B}),$$

where $\mathcal{B}$ fixes the admitted boundary variations and matching rule.

Theorem 9.2 (Bulk–boundary Ward balance). *For a boundary-complete source, the total diffeomorphism response has the form*

$$\mathcal{W}_{\text{tot}}(\mathcal{L}_X g) = -\langle J_{\text{bulk}}, X \rangle - \langle J_{\partial}, X|_{\partial M} \rangle.$$

The source is closed relative to the combined domain iff both declared residuals vanish after the matching rule is applied.

Proof. Apply the interior representation theorem and integration by parts separately to the bulk and boundary distributions. The matching rule fixes which normal flux is internal to the combined system and which is external. Closure is vanishing of the resulting total response for every admitted X . $\square$

9.2 Open and effective ledgers

Definition 9.3 (Directed exchange channel). A directed exchange channel is a named vector distribution J_{ν}^{ext} together with:

1. source and sink ledgers;
2. support and orientation;
3. continuation interval;

4. boundary or environment carrier;
5. reportable scope.

Proposition 9.4 (Exchange is not lawful equivalence). *A nonzero directed exchange channel need not possess an admissible inverse and may not be quotiented as presentation identity. It changes continuation content and must remain in the tensor/exchange ledger.*

Proof. Lawful equivalence preserves standing-bearing continuation. A directed exchange changes the source content of at least one subsystem. Unless the opposite channel is separately realized, no inverse exists. Therefore exchange is tensor-level continuation data, not skin. $\square$

9.3 Boundary omission obstruction

Theorem 9.5 (No hidden boundary repair). *Suppose a proposed bulk source has a nonzero boundary flux but is reported as closed without a boundary ledger. Then the report is source-incomplete or standing-failed. A later boundary term cannot convert the earlier incomplete act into the same closed source act.*

Proof. The nonzero boundary flux is a witness of source content leaving or entering the declared bulk domain. Omitting it makes the original source ledger incomplete. Adding it later may produce a corrected enlarged act, but irreversibility prevents that later act from changing the status of the original report. $\square$

9.4 Regularization and renormalization

A regulator, subtraction scheme, or finite improvement is skin only when it preserves the complete metric-response witness and Ward profile. If it changes either, it is standing-bearing source data or scope motion and must be re-evaluated. This is the source-side application of standing-preserving scale transport and quotient discipline [2, 5].

Proposition 9.6 (Renormalized source admission). *A renormalized source tensor has standing in this module only if:*

1. *the renormalized metric-response witness is defined on the complete probe space;*
2. *lawful scheme changes preserve its source quotient class;*
3. *local covariance is preserved;*
4. *the Ward residual is zero or declared;*
5. *state, domain, and boundary certificates are fixed.*

Proof. Each item is a coordinate of the source-bearing interface. Failure leaves the tensor scheme-dependent, noncovariant, incomplete, or unsupported by the claimed response class. $\square$

Part IV

Completed Composite Sources

10 Completed Composite Source Certificates

10.1 Why constituent addition is insufficient

A composite source is not determined by a list of constituent response classes. Binding, interaction, mediator-field, internal-stress, restoration, transport, vacuum/baseline, boundary, and failure terms may change the metric response. The module therefore does not define a binary operation

$$x \boxplus y$$

from constituents alone.

Definition 10.1 (Source term registry). A source term registry is a finite set $\mathcal{G}$ of uniquely identified standing-bearing source terms. Every term carries:

1. its upstream response ancestry;
2. its support and regularity class;
3. its upstream metric-attachment certificate and complete operational metric-response witness;
4. its continuation and Ward status;
5. its boundary and scale status.

Definition 10.2 (Sector family). A sector family is a finite set A . A sector may represent a constituent, binding term, interaction term, mediator field, internal stress, restoration term, transport ledger, vacuum/baseline term, or boundary term. Each sector $a \in A$ is assigned a subregistry $\mathcal{G}_a \subseteq \mathcal{G}$.

Definition 10.3 (No-double-counting cover). A sector cover is complete and nonduplicating when

$$\mathcal{G} = \bigsqcup_{a \in A} \mathcal{G}_a.$$

Equivalently, every registered source term has exactly one sector incidence.

Proposition 10.4 (No constituent-only composite). *If two completed composites share the same constituent response classes but differ in at least one witnessed binding, interaction, stress, restoration, boundary, or baseline term, they are distinct completed source presentations unless the differing term is source-equivalent.*

Proof. The differing term changes an operational metric-response witness or it does not. If it changes the witness, the complete source signature differs. If it does not, the difference is source-equivalent skin. No third role is available. $\square$

10.2 Sector witnesses

Every $\gamma \in \mathcal{G}_a$ is represented by a structured source-bearing triple $(p_\gamma, \mathbf{a}_\gamma, \mathbf{w}_\gamma)$. Define the sector witness

$$\mathcal{W}_a := \sum_{\gamma \in \mathcal{G}_a} \mathcal{W}_\gamma.$$

The sum is finite, so it is continuous and linear. Let $\mathbf{T}_a$ be its unique representing tensor distribution. The sector sum combines already attached witnesses; it does not attach a new functional to an unattached upstream response.

Definition 10.5 (Internal exchange channels). Let E be a finite set of internal exchange channels. Each channel $e \in E$ carries:

1. a vector distribution j_e ;
2. a support and continuation interval;
3. a signed incidence coefficient B_{ae} for every sector $a \in A$;
4. a proof or certificate that

$$\sum_{a \in A} B_{ae} = 0.$$

The matrix $B = (B_{ae})$ is the internal-exchange incidence matrix.

The coefficients need not be limited to ± 1 ; a hyperedge may distribute one channel among several sectors. Column balance expresses that an internal channel has no net source outside the completed composite.

Definition 10.6 (Sector Ward profile). A sector Ward profile is

$$\mathcal{W}_a(\mathcal{L}_X g) = -\langle J_a^{\text{ext}} + \sum_{e \in E} B_{ae} j_e, X \rangle$$

for every compactly supported vector field X , where J_a^{ext} is the declared external exchange of sector a .

Definition 10.7 (Completed composite source certificate). A completed composite source certificate is

$$\text{CompCert}(\mathcal{B}) = (\mathcal{G}, A, \{\mathcal{G}_a\}, \{\mathcal{W}_a\}, E, B, \{j_e\}, \{J_a^{\text{ext}}\}, \mathcal{K}, \mathcal{R}, \partial, \mathcal{F})$$

such that:

1. the term registry is finite;
2. the sector cover is complete and nonduplicating;
3. every registered term carries a complete source attachment and witness, and every sector witness is their finite sum;
4. every internal channel has balanced incidence;
5. every sector Ward profile holds;
6. $\mathcal{K}$ records support and overlap;

7. $\mathcal{R}$ records regularity;
8. ∂ records boundary matching;
9. $\mathcal{F}$ records anomaly, omission, and failure status.

Remark 10.8 (What the certificate does). The certificate does not generate binding or interaction physics. It requires every standing-bearing cross term actually used by the composite to be explicit, uniquely registered, metrically witnessed, and balance-typed. A missing term is not silently supplied by the word “composite”.

10.3 Certificate admission

Theorem 10.9 (Finite certificate decidability). *For a finite machine-readable certificate with explicit rational incidence coefficients and unique term identifiers, the following are decidable:*

1. registry coverage;
2. duplicate use;
3. unknown sector or channel references;
4. column balance of B ;
5. undeclared external residual labels;
6. closed versus open ledger status at the finite combinatorial level.

Proof. Each condition reduces to finite set comparison, identifier lookup, or finite exact arithmetic. The accompanying software object implements these checks. It does not decide analytic continuity, distributional existence, or the physical truth of a declared Ward profile. $\square$

11 Finite Internal-Exchange Cancellation

11.1 Total witness and total tensor

Definition 11.1 (Total composite witness). For a completed certificate, define

$$\mathcal{W}_{\mathcal{B}} := \sum_{a \in A} \mathcal{W}_a.$$

Theorem 11.2 (Total tensor construction). *The total witness is represented by*

$$\mathbf{T}_{\mathcal{B}} = \sum_{a \in A} \mathbf{T}_a.$$

Proof. For every $h \in \mathcal{V}_g$,

$$\mathcal{W}_{\mathcal{B}}(h) = \sum_a \mathcal{W}_a(h) = \frac{1}{2} \sum_a \langle \mathbf{T}_a, h \rangle = \frac{1}{2} \langle \sum_a \mathbf{T}_a, h \rangle.$$

Uniqueness in theorem 5.1 gives the result. $\square$

Corollary 11.3 (Composite support).

$$\text{supp } \mathbf{T}_B \subseteq \bigcup_{a \in A} \text{supp } \mathbf{T}_a.$$

Equality holds after removal of sectors whose tensor contributions cancel distributionally.

11.2 Cancellation theorem

Theorem 11.4 (Finite Internal-Exchange Cancellation). *For a completed composite certificate,*

$$\mathcal{W}_B(\mathcal{L}_X g) = -\left\langle \sum_{a \in A} J_a^{\text{ext}}, X \right\rangle$$

for every compactly supported vector field X . Equivalently,

$$\nabla_\mu \mathbf{T}_{B\nu}^\mu = J_{B,\nu}^{\text{ext}}, \quad J_B^{\text{ext}} := \sum_{a \in A} J_a^{\text{ext}}.$$

Proof. Sum the sector Ward profiles:

$$\begin{aligned} \mathcal{W}_B(\mathcal{L}_X g) &= -\sum_{a \in A} \langle J_a^{\text{ext}} + \sum_{e \in E} B_{ae} j_e, X \rangle \\ &= -\left\langle \sum_a J_a^{\text{ext}}, X \right\rangle - \sum_{e \in E} \left(\sum_{a \in A} B_{ae} \right) \langle j_e, X \rangle. \end{aligned}$$

Every internal column is balanced, so the second term vanishes. Apply theorem 8.2 to the total witness. $\square$

Corollary 11.5 (Closed completed composite). *If every $J_a^{\text{ext}} = 0$, then*

$$\nabla_\mu \mathbf{T}_{B\nu}^\mu = 0.$$

Corollary 11.6 (Declared-open completed composite). *If $J_B^{\text{ext}} \neq 0$, the exact total divergence equals the declared external exchange. Internal exchange does not contribute to the residual.*

11.3 Necessity of incidence balance

Theorem 11.7 (Unbalanced-channel obstruction). *Suppose an internal channel e_0 has*

$$c_{e_0} := \sum_{a \in A} B_{ae_0} \neq 0$$

and $j_{e_0} \neq 0$, with no equal and opposite declared channel. Then the total ledger has residual

$$c_{e_0} j_{e_0}$$

and is not closed.

Proof. The uncanceled term appears explicitly in the summed Ward profile. By theorem 8.2, it is part of the total divergence. $\square$

Corollary 11.8 (No verbal cancellation). *The statements “the interaction is internal,” “the exchange occurs once,” or “all terms have been listed” do not imply source closure. Closure requires the signed incidence balance or an equivalent proved Ward identity.*

11.4 Incidence-complex form

Let $\mathbb{R}^E$ be the channel space and $\mathbb{R}^A$ the sector space. The incidence matrix defines

$$B : \mathbb{R}^E \longrightarrow \mathbb{R}^A.$$

Let

$$\epsilon : \mathbb{R}^A \longrightarrow \mathbb{R}, \quad \epsilon((v_a)) = \sum_a v_a.$$

Column balance is the chain condition

$$\epsilon \circ B = 0.$$

Proposition 11.9 (Algebraic form of internal closure). *Internal exchange cancellation is precisely the statement that the totalization map ϵ annihilates the incidence image.*

Proof. For each basis channel e , $(\epsilon B)(e) = \sum_a B_{ae}$. □

This form is useful for mechanization and for generalized hypergraph composites.

12 Composite Source Closure and Failure Exhaustion

12.1 Main composite theorem

Theorem 12.1 (Composite Metric-Source Closure). *Let $\mathcal{B}$ possess a completed composite source certificate. Then:*

1. *the total operational witness exists and is continuous;*
2. *a unique total symmetric tensor distribution $\mathbf{T}_{\mathcal{B}}$ exists;*
3. *the total tensor is the finite sum of all uniquely registered sector tensors;*
4. *its support, naturality, and regularity follow from the corresponding sector certificates;*
5. *all certified internal exchanges cancel;*
6. *the exact divergence is the sum of declared external exchanges;*
7. *the composite is closed iff that external sum vanishes.*

Proof. Continuity follows from finite summation. Tensor existence and uniqueness follow from theorem 11.2. Support and regularity follow from finite distributional summation; naturality follows sectorwise and then by uniqueness. Internal cancellation and the divergence statement are theorem 11.4. The final equivalence is theorem 8.2. □

12.2 Action-presentable composites

Corollary 12.2 (Single-functional composite route). *Suppose the entire completed composite has one declared diffeomorphism-covariant functional*

$$\mathcal{S}_B = \sum_{a \in A} \mathcal{S}_a$$

whose term registry and field domains satisfy the no-double-counting cover. Then its Hilbert tensor is $\sum_a T_a$, and the covariant Noether identity supplies the sector Ward profiles and total closure on a certified solution.

Proof. Linearity of variation gives the total tensor. Diffeomorphism covariance gives the off-shell Ward identity. Evaluating on a certified solution and applying the declared boundary conditions gives the sector and total profiles. The incidence ledger records the decomposition of the already proved total identity. $\square$

Remark 12.3 (Why this does not assume every source has an action). The corollary is a sufficient route for the action-presentable subclass. The main composite theorem uses operational witnesses and remains valid without a primitive action.

12.3 Failure normal forms

Theorem 12.4 (Composite-source failure exhaustion). *A proposed completed composite that fails theorem 12.1 does so at one or more of the following first loci:*

1. **unrepresented term:** *a registered source term lacks a complete operational metric witness;*
2. **coverage gap:** *a standing-bearing term is omitted;*
3. **duplicate incidence:** *one term is counted more than once;*
4. **unbalanced internal channel:** *a column of B has nonzero sum;*
5. **undeclared external exchange:** *a nonzero external current is hidden;*
6. **boundary incompleteness:** *a boundary response or flux is omitted;*
7. **regularity mismatch:** *the tensor is promoted beyond its certificate;*
8. **anomaly:** *the Ward residual is not generated by declared exchange;*
9. **off-shell promotion:** *a formal identity is reported as on-shell closure without a solution certificate;*
10. **scope motion or standing failure.**

Proof. The composite theorem uses exactly: complete term coverage, unique incidence, represented sector witnesses, balanced internal exchange, declared external and boundary status, typed regularity, and a valid Ward profile. Failure of the conclusion therefore has a first failure among those inputs or changes the target. $\square$

Proposition 12.5 (No repair by post-hoc tensor addition). *If a composite fails because a source term or exchange channel was omitted, adding that term later constructs a new corrected composite certificate. It does not make the earlier incomplete certificate closed as the same act.*

Proof. The earlier certificate has a fixed registry and act-time status. A later addition changes the registry and source object. Irreversibility distinguishes the corrected act from the failed original. $\square$

12.4 Interface to the Einstein source ledger

Corollary 12.6 (Smooth closed source interface). *If every sector has a smooth density certificate, the completed composite is closed, and the source system meets the declared Einstein Cauchy regularity assumptions, then*

$$T_{\mu\nu}^{\mathcal{B}} \in \text{Src}_g^{\text{sm}}, \quad \nabla^\mu T_{\mu\nu}^{\mathcal{B}} = 0.$$

It is therefore eligible for the smooth closed source coordinate of the minimal Einstein metric projection.

Proof. Finite sums preserve smoothness. Closure is corollary 11.5. Eligibility for the Einstein source coordinate is then a type statement; this module does not prove the coupled Einstein Cauchy development. $\square$

Part V

Inhabited Models and Audit Artifact

13 Inhabited Analytic Models

The examples prove that the module interfaces are nonempty. They do not replace the class-wide attachment, representation, and closure theorems.

13.1 A standing response with no metric attachment

Example 13.1 (Finite non-source-bearing response model). Let p_{int} have one admitted upstream protocol π_{int} with a nonzero internal response output, but let the fixed source-projection envelope contain no protocol whose faithful metric tangent lies in $\mathcal{V}_g$. Thus

$$\text{Prot}_{p_{\text{int}}} = \{\pi_{\text{int}}\}, \quad \text{Out}_{p_{\text{int}}}(\pi_{\text{int}}) = 1, \quad \text{MetInc}_{p_{\text{int}}} = \emptyset.$$

The presentation has a nontrivial upstream response class in Q_{resp} , but totality of a metric-attachment certificate fails. Hence

$$[p_{\text{int}}]_{\sim_{\text{resp}}} \notin Q_{\text{resp}}^{\text{src}}.$$

This finite witness model proves that the source-bearing subtype is not obtained by pairing every upstream response with the zero functional.

13.2 Certified zero metric response

Example 13.2 (Attached zero witness). Let p_0 possess a complete metric-incidence protocol family, and suppose the upstream protocol ledger proves

$$\text{Out}_{p_0}(\pi) = 0 \quad \text{for every } \pi \text{ incident to every } h \in \mathcal{V}_g.$$

Then $\text{MetResp}_{p_0} = \mathcal{V}_g \times \{0\}$, so corollary 3.10 gives a genuine attachment of the zero functional. The resulting tensor distribution is zero. This is source-neutrality proved by the upstream response protocol, not an arbitrary zero lift.

13.3 Smooth real scalar field

Let (M, g) be a smooth Lorentzian manifold and let $\phi \in C^\infty(M)$ have the declared support and boundary behavior. Consider

$$\mathcal{S}_\phi[g, \phi] = -\frac{1}{2} \int_M \left(g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi + m^2 \phi^2 \right) \text{dvol}_g.$$

Proposition 13.3 (Scalar-field witness). *The scalar response presentation includes the finite metric-deformation protocol $g \mapsto \mathcal{S}_\phi[g, \phi]$. Its Fréchet derivative graph is the upstream metric-response relation, so proposition 4.5 supplies an attachment certificate and a complete smooth source witness represented by*

$$T_{\mu\nu}^\phi = \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} \left(\nabla_\alpha \phi \nabla^\alpha \phi + m^2 \phi^2 \right).$$

Moreover,

$$\nabla^\mu T_{\mu\nu}^\phi = (\square_g \phi - m^2 \phi) \nabla_\nu \phi$$

up to the fixed sign convention for the Klein–Gordon equation. Hence every certified solution is closed.

Proof. Differentiate the action with respect to the metric to obtain the Hilbert tensor. Direct covariant differentiation and cancellation of the mixed second-derivative terms gives the displayed identity. On the Klein–Gordon solution class the right-hand side vanishes. $\square$

This example inhabits $\widehat{Q}_{\text{src}}^{\text{sm}}$ and the closed smooth Einstein-source interface.

13.4 Distributional point body

Let $\gamma : I \rightarrow M$ be a smooth timelike worldline with unit tangent u^μ and let $m > 0$. Define

$$\mathcal{W}_\gamma(h) = \frac{m}{2} \int_I u^\mu u^\nu h_{\mu\nu}(\gamma(\tau)) d\tau$$

for compactly supported h .

Proposition 13.4 (Worldline source distribution). *The worldline response presentation declares metric deformation of the proper-time functional as its upstream metric-incidence protocol. Its directional derivative is continuous linear and graph-complete, so the witness is attached and represented by the symmetric tensor distribution*

$$\mathbf{T}_\gamma^{\mu\nu} = m \int_I u^\mu u^\nu \delta_{\gamma(\tau)} d\tau.$$

For compactly supported vector fields away from worldline endpoints,

$$\nabla_\mu \mathbf{T}_\gamma^\mu = m \int_I a_\nu(\tau) \delta_{\gamma(\tau)} d\tau,$$

where $a^\nu = u^\mu \nabla_\mu u^\nu$.

Proof. Continuity follows from restriction of test sections to the smooth curve and the compact support intersection. The representation formula is immediate. Pair the divergence with X , integrate $u^\mu \nabla_\mu (X_\nu u^\nu)$ by parts, and use compact support to remove endpoint terms. The residual is the acceleration distribution. $\square$

Corollary 13.5 (Worldline Ward classification). *The worldline source is closed precisely when γ is geodesic in the declared free-body scope. A force gives an exchange current rather than a hidden failure.*

This example shows why the module retains distributional sources and why a Ward certificate is stronger than merely writing a familiar tensor formula.

13.5 Two-sector internal exchange

Let sectors A and B satisfy

$$\nabla_\mu \mathbf{T}_{A^\nu}^\mu = j_\nu + k_\nu, \quad \nabla_\mu \mathbf{T}_{B^\nu}^\mu = -j_\nu,$$

where j is internal and k is external. The incidence matrix for the internal channel is

$$B = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Proposition 13.6 (Two-sector closure).

$$\nabla_\mu (\mathbf{T}_A + \mathbf{T}_B)^\mu{}_\nu = k_\nu.$$

Thus the composite is closed when $k = 0$ and otherwise has exactly the declared external exchange.

Proof. The internal channel column sums to zero. Apply theorem 11.4. $\square$

13.6 Interaction-functional realization

Let ϕ, χ be smooth scalar fields with total action

$$\mathcal{S}[g, \phi, \chi] = \int_M \sqrt{-g} \left(-\frac{1}{2} |\nabla \phi|^2 - \frac{1}{2} |\nabla \chi|^2 - V(\phi, \chi) \right) d^n x.$$

The interaction potential occurs once in the source registry.

Proposition 13.7 (Single-functional interaction closure). *The total Hilbert tensor is the sum of the two kinetic sector tensors and the one interaction tensor. On a certified joint solution,*

$$\nabla^\mu T_{\mu\nu}^{\text{tot}} = 0.$$

Any sector split produces equal-and-opposite internal exchange whose incidence sum vanishes.

Proof. Apply corollary 12.2. The joint field equations cancel the interaction derivatives in the total divergence. A sector decomposition records that cancellation as an internal incidence channel. $\square$

14 Machine-Readable Composite Ledger Checker

The project includes an executable Python package, `source_ledger_checker`. It validates the finite combinatorial coordinates of definition 10.7.

14.1 Verified properties

Given a JSON certificate, the checker verifies:

1. unique sector, term, and channel identifiers;
2. exact term coverage and absence of duplicate term incidence;
3. validity of every sector and channel reference;
4. exact rational column balance of the internal-exchange incidence matrix;
5. declaration of external exchange labels;
6. closed/open status from the finite residual ledger;
7. deterministic canonical serialization and SHA-256 identity.

Theorem 14.1 (Software theorem correspondence). *If the checker accepts a certificate as internally balanced, then the finite incidence hypothesis*

$$\sum_{a \in A} B_{ae} = 0 \quad (e \in E)$$

and the no-double-counting term-cover hypothesis used by theorem 11.4 hold for the parsed certificate.

Proof. The parser constructs the exact finite sets and rational incidence coefficients from the JSON object. Acceptance requires equality of the term registry with the disjoint union of sector term lists and exact zero column sums using rational arithmetic. These are precisely the finite hypotheses. $\square$

The software does not verify:

- upstream metric-incidence or source attachment;
- continuity of metric-response functionals;
- existence of tensor distributions;
- diffeomorphism covariance;
- correctness of analytic Ward profiles;
- physical adequacy of declared sectors or currents;
- PDE existence or regularity.

Those burdens remain in the manuscript proof and source certificate.

14.2 Command-line use

From the project root:

```
python -m source_ledger_checker check \
    software/examples/closed_two_sector.json
```

```
python -m source_ledger_checker check \
    software/examples/open_three_sector.json
```

```
python -m unittest discover -s software/tests -v
```

The closed example returns a zero internal residual. The open example returns only the explicitly declared external current. Deliberately unbalanced and duplicate-term fixtures are rejected by the tests.

14.3 Audit role

The software is a finite certificate validator, not proof authority for the analytic theorems. Its purpose is to prevent transcription errors, hidden duplicate terms, and false incidence cancellation when a completed composite ledger is later assembled for the capstone.

Part VI

Endpoint, Handoff, and Adversarial Closure

15 Main Module Theorems

Theorem 15.1 (Module A Source-Admission and Attachment Theorem). *Fix the kernel-governed realized physical interior and a faithful metric-projection envelope. Then:*

1. *an upstream response class belongs to the base source-bearing subtype $Q_{\text{resp}}^{\text{src}} \subseteq Q_{\text{resp}}$ exactly when it carries a complete upstream metric-attachment certificate and a complete operational metric witness;*
2. *every complete attachment certificate determines a unique continuous linear metric-response functional whose graph is the upstream metric-response relation;*
3. *the zero functional attaches exactly when the complete upstream metric-incidence protocol family certifies zero response on every test variation;*
4. *the structured quotient $\hat{Q}_{\text{src}}$ retains the attachment certificate and complete witness and maps surjectively to $Q_{\text{resp}}^{\text{src}}$.*

Proof. The unique attached functional is theorem 3.8; the zero criterion is corollary 3.10; invariance and the two source spaces are proposition 3.12 and definitions 3.15 and 3.17. The non-tautological boundary is exhibited by proposition 3.19 and example 13.1. $\square$

Theorem 15.2 (Module A Source-Representation Theorem). *For every structured source-witness class*

$$x = [(p, \mathfrak{a}_p, \mathfrak{w}_p)] \in \widehat{Q}_{\text{src}}$$

there exists a unique lawful class of symmetric tensor-density distributions

$$q_g(x) = [\mathbf{T}_x] \in \text{Src}_g^{\text{dist}} / \simeq_{\text{law}}$$

such that

$$\mathcal{W}_x(h) = \frac{1}{2} \langle \mathbf{T}_x, h \rangle \quad (h \in \mathcal{V}_g).$$

The map q_g is well defined, natural under lawful metric redescription, support-preserving, regularity-typed, and faithful relative to the complete metric probe quotient.

Proof. Attachment has already fixed which operational functional belongs to the upstream response. Existence and uniqueness of its tensor dual are theorem 5.1. Naturality is theorem 6.1; support is theorem 5.4; regularity typing is proposition 5.9 and theorem 7.2; quotient descent and faithfulness are theorems 6.4 and 6.5. $\square$

Theorem 15.3 (Module A Ward-Closure Theorem). *For every $x \in \widehat{Q}_{\text{src}}$ with declared Ward/exchange current J_x ,*

$$\nabla_\mu \mathbf{T}_{x\nu}^\mu = J_{x,\nu}.$$

Hence:

1. $J_x = 0$ iff the source is closed;
2. $J_x \neq 0$ with a named ledger is an exact open/effective/boundary exchange;
3. $J_x \neq 0$ without a named ledger is anomaly, omission, off-shell status, scope motion, or standing failure.

Proof. Apply theorem 8.2 and proposition 8.8. $\square$

Theorem 15.4 (Module A Composite-Closure Theorem). *Every completed composite source certificate determines a unique total source tensor distribution. Every internal exchange cancels exactly by finite incidence balance, and*

$$\nabla_\mu \mathbf{T}_{\mathcal{B}\nu}^\mu = \sum_{a \in A} J_{a,\nu}^{\text{ext}}.$$

The composite is closed iff the declared external sum vanishes.

Proof. Use theorem 12.1. Each component source used by the composite predicate is already attachment-certified by definition 3.14. $\square$

Theorem 15.5 (Operational–Hilbert Compatibility). *On the attached action-presentable subclass, the operationally represented source tensor is exactly the Hilbert metric variation of the declared source functional. The operational construction is therefore compatible with conventional stress-energy extraction without making action primitivity a premise.*

Proof. Use theorem 7.5 and proposition 4.5. □

Theorem 15.6 (Module Endpoint Closure). *The six frozen burdens of AASC – GRQM – MA – SRC – 1.0 are discharged:*

1. *the base source-bearing subtype $Q_{\text{resp}}^{\text{src}}$ and the structured quotient $\widehat{Q}_{\text{src}}$ are exactly distinguished and attachment-safe;*
2. *tensor representation exists and is unique for every attached complete witness;*
3. *support, naturality, regularity, and quotient descent are proved;*
4. *Ward/divergence equivalence is proved;*
5. *finite composite closure is constructed and proved;*
6. *the analytic and machine-readable interfaces are nonempty and independently auditable.*

Proof. The first item is section 3 and theorem 15.1; items two and three are sections 5 to 7; item four is sections 8 and 9; item five is sections 10 to 12; item six is sections 13 and 14 and the passing test suite included in the project. □

15.1 Exact strength of the endpoint

The endpoint is stronger than the statement “a stress-energy tensor may be written” because it first proves which upstream responses possess an attached metric first-response witness, then constructs the tensor from that witness, proves quotient descent, derives the divergence profile, and certifies finite composite cancellation. It is weaker than a theorem that every RPI response is source-bearing, that every base source class has a unique attached structure, or that every source class has a smooth local action. Those distinctions are deliberate and fixed.

16 Capstone Handoff and Dependency Interface

This module is not integrated into the capstone in the present release. It supplies an independently reviewable theorem object for later use.

16.1 Objects exported

The module exports:

1. the attachment relation Attach_g , the true base subtype $Q_{\text{resp}}^{\text{src}} \subseteq Q_{\text{resp}}$, and the structured source quotient $\widehat{Q}_{\text{src}}$;
2. the natural quotient-faithful source map

$$q_g : \widehat{Q}_{\text{src}} \rightarrow \text{Src}_g^{\text{dist}} / \simeq_{\text{law}};$$

3. the Ward/exchange identity

$$\text{div } q_g(x) = J_x;$$

4. the completed composite constructor and internal-exchange cancellation theorem;

5. the smooth-interface structured subdomain $\widehat{Q}_{\text{src}}^{\text{sm}}$;
6. the finite certificate schema and validator.

16.2 Later overlap with inertial response

A later charge/matching module may define the base overlap

$$Q_{\text{mass}} := Q_{\text{in}} \cap Q_{\text{resp}}^{\text{src}},$$

where Q_{in} is the independently constructed inertial transport subtype. The metric source map is still evaluated on the structured source class lying above that base point. This manuscript does not identify the inertial and source structures and does not prove a universal ratio between their numerical readouts.

16.3 Gate contribution

Relative to the frozen capstone contract, the accepted form of this module is intended to supply:

- the attachment-safe metric-source half of common-response factorization;
- composite source closure and exact declared exchange;
- source, provenance, and regularity data needed by later continuation and occupancy modules.

It does not close:

- uniqueness of source attachment over a fixed base response class;
- the Hamiltonian/inertial map;
- the symmetry/charge/nonrelativistic contraction bridge;
- the upstream continuation and projection intertwining theorem;
- fibre-by-fibre certificate uniqueness;
- the final assignment orbit.

16.4 Import rule

A later manuscript may cite theorems 15.1 to 15.4 only after fixing:

1. the upstream response class and its complete attachment certificate;
2. the exact structured source-witness class;
3. the metric envelope;
4. the regularity required by the later theorem;
5. closed or exchange-declared status;
6. all composite and boundary certificates actually used.

The phrase “Module A supplies a source tensor” is insufficient without those coordinates.

17 Adversarial Map and Same-Scope Attack Closure

Every hostile objection should be tested first against the exact theorem chain below. An objection already answered here has standing only if it identifies the precise premise or inference that fails.

Objection	Standing?	Exact response
The paper assumes a stress tensor.	No.	The attached operational metric-response witness is the input. The tensor distribution is constructed uniquely by theorem 5.1.
The paper defines source-bearing so the result is tautological.	No, unless the attachment theorem fails.	The base subtype is selected by a complete upstream metric-incidence certificate. The operational functional must have graph exactly equal to the response relation already determined by that protocol ledger. The finite model in example 13.1 lies in Q_{resp} but not $Q_{\text{resp}}^{\text{src}}$.
The zero functional gives every response a source-bearing lift.	No.	By corollary 3.10, zero attaches iff every protocol in a total complete metric-incidence family has upstream output zero. A response lacking metric protocols fails totality; an arbitrary zero ledger fails graph equality.
The structured source space is not literally a subtype of Q_{resp} .	Correct and repaired.	The true subtype is $Q_{\text{resp}}^{\text{src}} \subseteq Q_{\text{resp}}$. The domain of q_g is the structured quotient $\hat{Q}_{\text{src}}$, with a surjective forgetful map to the base subtype. Distinct attached source structures over one base response remain distinct pending a later fibre theorem.
The witness is secretly an action derivative.	No.	Action presentation is optional. A direct ledger is allowed only after graph-equality attachment; theorem 7.5 proves compatibility on the action-presentable subclass.
A Gâteaux derivative need not be continuous linear.	Correct in general and explicitly controlled.	Definition 4.4 requires Fréchet/Hadamard differentiability on the chosen locally convex model, or a separately certified additive, homogeneous, LF-continuous Gâteaux differential.
Distributional duality is merely notation.	No.	It supplies existence and uniqueness in the exact tensor bundle, support identity, regularity stratification, covariance, and the Ward/divergence bridge.
Stress tensors have improvement ambiguity.	Only if the probe family is restricted or boundary data are omitted.	All compact interior metric variations are used. A distribution pairing to zero with all such probes is zero; boundary-supported differences are separately typed.

Objection	Standing?	Exact response
Conservation is assumed in the Ward certificate.	No for the operational theorem; yes if a model merely labels itself closed.	The module distinguishes an operational Ward profile from a claim of closure and proves its exact equivalence to tensor divergence. Action models must additionally supply an on-shell solution certificate.
Internal exchanges cancel because the paper says they are internal.	No.	Cancellation is a finite signed-incidence theorem. Unbalanced channels produce an explicit residual and are rejected.
Listing all composite terms proves closure.	No.	Coverage and no-duplication are separate from balance and Ward status. All are required by definition 10.7.
The software proves the physics.	No.	The software verifies only the finite registry and incidence hypotheses. Analytic attachment, continuity, covariance, and Ward identities remain manuscript obligations.
A point-particle example proves every source exists.	No.	Examples prove nonemptiness only. Source admission is class-wide only on the base subtype selected by theorem 3.8; tensor representation is class-wide only on the structured quotient $\hat{Q}_{\text{src}}$.
The module should prove universal mass equivalence.	Proof-class mismatch.	That is Module B. This module exports q_g on $\hat{Q}_{\text{src}}$ and does not identify the inertial overlap.
The module uses metricity to generate response standing.	No.	The kernel, RPI, particle, and deformation-response roles are upstream. Theorem 1.3 fixes the order.

17.1 Correctness-relevant denial targets

A same-mode criticism is correctness-relevant if it shows one of the following:

1. the upstream metric-incidence relation is not independent of the proposed witness;
2. totality, coherence, first-order linearity, continuity, lawful transport, or protocol completeness in definition 3.7 fails;
3. an attached functional does not have graph equal to the upstream metric-response relation;
4. the base subtype is not invariant under upstream response equivalence;
5. the operational witness is not continuous or linear on the stated test space;
6. the dual bundle in theorem 5.1 is incorrect;
7. the complete-probe uniqueness inference fails;
8. source equivalence does not make q_g well defined;

9. naturality does not transport the tensor distribution;
10. the Lie-derivative pairing has the wrong divergence identity;
11. the composite term cover is incomplete or duplicating;
12. the incidence cancellation theorem is algebraically false;
13. the software does not implement the stated finite hypotheses;
14. the manuscript reports beyond its explicit nonclaims.

Framework preference, demand for a primitive matter action, or demand that the module solve the later capstone gates does not by itself refute the stated endpoint.

18 Conclusion

The source side of the Einstein projection cannot be licensed by writing a familiar stress-energy tensor, by attaching an arbitrary functional to an upstream response, or by generating source standing backward from the Einstein equation. The repaired dependency order is:

$$\text{TargetAdeq}_D \implies K_D \implies A_D^+ \implies I_D \implies Q_{\text{resp}} \implies Q_{\text{resp}}^{\text{src}} \longleftarrow \hat{Q}_{\text{src}} \xrightarrow{q_g} q_g(\hat{Q}_{\text{src}}).$$

The first new theorem is the attachment theorem. The complete upstream response signature already contains the admitted protocols, their metric-incidence relation, and their outputs. Totality, coherence, first-order vector compatibility, continuity, lawful transport, and protocol completeness make that relation the graph of one continuous linear functional. An arbitrary functional, including an arbitrary zero functional, does not attach. Zero is admitted only when the complete upstream metric-response protocol certifies zero output for every metric test.

This construction separates the true base source-bearing subtype $Q_{\text{resp}}^{\text{src}} \subseteq Q_{\text{resp}}$ from the structured source-witness quotient $\hat{Q}_{\text{src}}$. The latter retains the attachment certificate and complete witness and is the correct domain of the tensor representation map. Multiple inequivalent source structures may lie over one base response class; Module A does not erase that later fibre burden.

Distributional duality then produces one and only one symmetric tensor distribution from the attached operational response. The complete probe family removes local tensor ambiguity; lawful-redescription naturality descends the tensor through the source quotient; regularity remains explicitly typed rather than silently promoted.

The Ward coordinate of the operational witness is exactly the distributional divergence of the represented tensor. Closed source continuation is therefore not an imported GR conservation postulate. It is the metric form of vanishing response to compactly supported diffeomorphism directions. Nonzero response is retained as a directed external, boundary, anomalous, or effective exchange rather than laundered into equivalence.

Composite closure requires more than constituent addition. The completed certificate records every binding, interaction, field, stress, restoration, transport, baseline, and boundary term exactly once. A signed incidence complex then proves cancellation of internal exchange and exposes every residual. The software artifact independently checks the finite combinatorial hypotheses while leaving analytic attachment and representation burdens visible.

The compressed endpoint is:

$$\begin{aligned}
&\text{complete upstream metric-incidence certificate} \iff \text{unique attached operational witness,} \\
&\text{attached complete witness} \iff \text{unique symmetric tensor distribution,} \\
&\mathcal{W}(\mathcal{L}_X g) = -\langle J, X \rangle \iff \nabla_\mu \mathbf{T}^\mu{}_\nu = J_\nu, \\
&\text{balanced completed composite} \implies \nabla_\mu \mathbf{T}^\mu{}_{\text{tot}\nu} = J_\nu^{\text{ext}}.
\end{aligned}$$

This closes the bounded Cycle-1 defect without amending the frozen Module A endpoint and without claiming the later symmetry/charge, upstream continuation, source-fibre uniqueness, or capstone-orbit results.

A Formal Predicates and Type Conditions

A.1 Kernel and projection order

$$\text{TargetAdeq}_D(T) \implies K_D \implies A_D^+ \implies I_D.$$

A faithful metric projection satisfies:

$$\begin{aligned}
\text{FaithfulMetricProj}_D(\Pi_g) \iff & \text{Downstream}(\Pi_g) \wedge \text{AnchorPres}(\Pi_g) \wedge \text{TensorPres}(\Pi_g) \\
& \wedge \text{SkinKernel}(\Pi_g) \wedge \text{QuotPres}(\Pi_g) \wedge \text{ContPres}(\Pi_g) \wedge \text{NoBackflow}(\Pi_g).
\end{aligned}$$

A.2 Upstream metric-response relation

For $p \in \text{RespPres}_D$, let $\Pi_g : \mathcal{P}_p \rightarrow \text{Met}(M)$ be the fixed faithful metric projection. For an admitted C^1 response path γ_δ representing δ , define

$$\tau_{g,p}(\delta) := \left. \frac{d}{ds} \right|_{s=0} \Pi_g(\gamma_\delta(s)) \in \mathcal{V}_g,$$

when the derivative exists, is compactly supported, and is independent of the admitted path representative modulo projection skin. This gives the partial map $\tau_{g,p} : \Delta_p^{\text{adm}} \rightarrow \mathcal{V}_g$. Define

$$\text{MetInc}_p(h, \pi) \iff \exists \delta \in \Delta_p^{\text{adm}} [\text{Inc}_p(\pi, \delta) \wedge \tau_{g,p}(\delta) = h].$$

Then

$$\text{MetResp}_p(h, a) \iff \exists \pi \in \text{Prot}_p [\text{MetInc}_p(h, \pi) \wedge \text{Out}_p(\pi) = a].$$

A complete attachment certificate satisfies

$$\begin{aligned} \text{AttCert}_g(p, \mathfrak{a}) \iff & \text{TangentDerived}(p, \tau_{g,p}, \mathfrak{a}) \\ & \wedge \text{UpstreamProvenance}(p, \mathfrak{a}) \\ & \wedge \text{Total}(\text{MetResp}_p) \\ & \wedge \text{SingleValued}(\text{MetResp}_p) \\ & \wedge \text{FirstOrderLinear}(\text{MetResp}_p) \\ & \wedge \text{LFContinuous}(\text{MetResp}_p) \\ & \wedge \text{LawfulTransport}(\mathfrak{a}) \\ & \wedge \text{MetricProtocolComplete}(p, \mathfrak{a}). \end{aligned}$$

The attachment relation is

$$\text{Attach}_g(p, \mathfrak{a}, \mathcal{W}) \iff \text{AttCert}_g(p, \mathfrak{a}) \wedge \text{Graph}(\mathcal{W}) = \text{MetResp}_p.$$

Thus

$$\text{Attach}_g(p, \mathfrak{a}, 0) \iff \text{AttCert}_g(p, \mathfrak{a}) \wedge \forall h, \pi [\text{MetInc}_p(h, \pi) \Rightarrow \text{Out}_p(\pi) = 0].$$

The zero functional is therefore not a universal lift.

A.3 Source-bearing predicate and two source spaces

For $p \in \text{RespPres}_D$,

$$\begin{aligned} \text{CompleteMetricWitness}_g(\mathfrak{w}) \iff & \mathfrak{w} = (\mathcal{W}, K, \text{Nat}, J, \text{Reg}, \partial) \\ & \wedge \mathcal{W} \in \mathcal{L}_{\text{cont}}(\mathcal{V}_g, \mathbb{R}) \wedge \text{supp } \mathcal{W} \subseteq K \\ & \wedge \text{Natural}(\mathcal{W}, \text{Nat}) \wedge \forall X \in \Gamma_c(TM), \mathcal{W}(\mathcal{L}_X g) = -\langle J, X \rangle \\ & \wedge \text{TypedRegularity}(\text{Reg}) \wedge \text{TypedBoundary}(\partial). \end{aligned}$$

Then

$$\text{SrcBear}_g(p, \mathfrak{a}, \mathfrak{w}) \iff \text{Attach}_g(p, \mathfrak{a}, \mathcal{W}_{\mathfrak{w}}) \wedge \text{CompleteMetricWitness}_g(\mathfrak{w}).$$

The true base subtype is

$$Q_{\text{resp}}^{\text{src}} = \{[p]_{\sim_{\text{resp}}} \in Q_{\text{resp}} : \exists \mathfrak{a}, \mathfrak{w} \text{ SrcBear}_g(p, \mathfrak{a}, \mathfrak{w})\} \subseteq Q_{\text{resp}}.$$

The structured source-witness quotient is

$$\widehat{Q}_{\text{src}} = \{(p, \mathfrak{a}, \mathfrak{w}) : \text{SrcBear}_g(p, \mathfrak{a}, \mathfrak{w})\} / \sim_{\text{src}},$$

with a surjective forgetful map

$$F : \widehat{Q}_{\text{src}} \twoheadrightarrow Q_{\text{resp}}^{\text{src}}.$$

The metric-source map is defined on $\widehat{Q}_{\text{src}}$, because its value depends on the complete attached witness, not merely on the underlying response class.

A.4 Representation map

$$\mathcal{R}_g : \mathcal{L}_{\text{cont}}(\mathcal{V}_g, \mathbb{R}) \xrightarrow{\sim} \mathcal{D}'(S^2TM \otimes |\Lambda|),$$

defined by

$$\mathcal{W}(h) = \frac{1}{2} \langle \mathcal{R}_g(\mathcal{W}), h \rangle.$$

Then

$$q_g([(p, \mathfrak{a}, \mathfrak{w})]) = [\mathcal{R}_g(\mathcal{W}_p)]_{\simeq_{\text{law}}}.$$

A.5 Ward predicate

$$\text{Ward}_g(\mathcal{W}, J) \iff \forall X \in \Gamma_c(TM), \mathcal{W}(\mathcal{L}_X g) = -\langle J, X \rangle.$$

The theorem is:

$$\text{Ward}_g(\mathcal{W}, J) \iff \text{div } \mathcal{R}_g(\mathcal{W}) = J.$$

A.6 Composite certificate predicate

For

$$C = (\mathcal{G}, A, \{\mathcal{G}_a\}, \{(p_a, \mathfrak{a}_a, \mathfrak{w}_a)\}, E, B, \{j_e\}, \{J_a^{\text{ext}}\}, \mathcal{K}, \mathcal{R}, \partial, \mathcal{F}),$$

write $\text{CompleteComp}(C)$ iff:

$$\begin{aligned} & |\mathcal{G}|, |A|, |E| < \infty, \\ & \mathcal{G} = \bigsqcup_{a \in A} \mathcal{G}_a, \\ & \forall \gamma \in \mathcal{G}, \text{SrcBear}_g(p_\gamma, \mathfrak{a}_\gamma, \mathfrak{w}_\gamma), \\ & \forall a, \mathcal{W}_a = \sum_{\gamma \in \mathcal{G}_a} \mathcal{W}_\gamma, \\ & \forall e, \sum_a B_{ae} = 0, \\ & \forall a, X, \mathcal{W}_a(\mathcal{L}_X g) = -\langle J_a^{\text{ext}} + \sum_e B_{ae} j_e, X \rangle, \\ & \text{TypedSupportBoundaryRegularityFailure}(C). \end{aligned}$$

Then:

$$\text{CompleteComp}(C) \implies \text{div } \mathbf{T}_C = \sum_a J_a^{\text{ext}}.$$

B Certificate and JSON Schema

B.1 Analytic certificate coordinates

Coordinate	Data	Proof burden
Response ancestry	Upstream response identifiers and quotient class	No metric or tensor backflow; response class fixed before attachment.

Coordinate	Data	Proof burden
Metric attachment	Upstream protocol identifiers, metric-incidence relation, protocol outputs, composition/rescaling and completeness certificates	Graph equality with the attached functional; arbitrary zero and arbitrary fitted witnesses excluded.
Metric envelope	M, g, ∇ , dimension, signature, boundary class	Fixed projection anchors and lawful redescription class.
Probe space	$\Gamma_c(S^2T^*M)$	Completeness and topology.
Witness	Continuous linear $\mathcal{W}$	Must be the unique functional whose graph equals the upstream metric-response relation.
Support	Closed support class K	Vanishing on disjoint probes.
Naturality	Diffeomorphism transport law	Equality of transported witness.
Ward/exchange	Vector distribution J	Equality on every $\mathcal{L}_X g$.
Regularity	Distribution, measure, local integrability, Sobolev, or smooth	No promotion beyond certificate.
Boundary	Interior or bulk-boundary pair	Flux and matching rule.
Failure status	Closed, open, anomalous, or off shell	Exact reportability.

B.2 Finite composite JSON object

The software schema uses the following top-level fields:

```
{
  "module": "AASC-GRQM-MA-SRC-1.0",
  "certificate_id": "...",
  "terms": [{"id": "...", "kind": "..."}],
  "sectors": [
    {
      "id": "...",
      "terms": ["..."],
      "external_exchange": ["..."]
    }
  ],
  "channels": [
    {
      "id": "...",
      "current": "...",
      "incidence": {"sector_A": "1", "sector_B": "-1"}
    }
  ],
  "declared_external_currents": ["..."],
  "boundary_status": "closed|declared",
  "notes": "..."
}
```

Incidence values are parsed as exact rational numbers.

B.3 Acceptance conditions

The checker accepts a certificate when:

1. all identifiers are unique;
2. every term occurs in exactly one sector;
3. every referenced sector exists;
4. every internal channel column sums to zero;
5. every external label is declared;
6. the requested closed/open report matches the residual ledger.

An accepted JSON file certifies only these finite conditions. The analytic source certificate remains mandatory.

C Expanded Proof Details

C.1 Dual bundle calculation

For $E = S^2T^*M$,

$$E^* \cong S^2TM.$$

The continuous dual of compactly supported smooth sections of E is

$$\Gamma_c(E)' = \mathcal{D}'(E^* \otimes |\Lambda|M).$$

The density factor makes the pairing coordinate independent without choosing a volume form. Once g is fixed, dvol_g identifies density-valued distributions with tensor distributions in the conventional notation.

C.2 Distributional divergence sign

For smooth compactly supported X ,

$$\begin{aligned} \mathcal{W}(\mathcal{L}_X g) &= \frac{1}{2} \langle \mathbf{T}^{\mu\nu}, \nabla_\mu X_\nu + \nabla_\nu X_\mu \rangle \\ &= \langle \mathbf{T}^{\mu\nu}, \nabla_\mu X_\nu \rangle \\ &= -\langle \nabla_\mu \mathbf{T}^\mu{}_\nu, X^\nu \rangle. \end{aligned}$$

Thus the manuscript's sign convention is

$$\mathcal{W}(\mathcal{L}_X g) = -\langle J, X \rangle \iff \text{div } \mathbf{T} = J.$$

C.3 Composite cancellation in index form

Given

$$\text{div } \mathbf{T}_a = J_a^{\text{ext}} + \sum_e B_{ae} j_e,$$

finite summation gives

$$\begin{aligned} \text{div } \mathbf{T}_{\text{tot}} &= \sum_a J_a^{\text{ext}} + \sum_e \left(\sum_a B_{ae} \right) j_e \\ &= \sum_a J_a^{\text{ext}}. \end{aligned}$$

No commutation of infinite sums, weak limits, or nonlinear operations is used.

C.4 Regularity warning

A derivative-of-delta source has finite distribution order but no measure density. A worldline source is a tensor-valued measure supported on a curve but is not L^1_{loc} relative to spacetime volume. A smooth classical field source belongs to Src_g^{sm} . These classes must not be conflated when importing the module into an Einstein existence theorem.

C.5 Attachment, completeness, and non-tautology

The source-admission theorem has two distinct inputs and outputs.

First, the upstream response presentation already carries a complete protocol signature and a faithful metric tangent map. These determine the relation MetResp_p . The attachment theorem proves that, when the relation is total, coherent, first-order linear, continuous, natural, and protocol-complete, it is the graph of one continuous linear functional. No prospective tensor or functional determines that graph. A response with no metric-incidence protocols therefore fails source admission, while the zero functional attaches only when the complete relation is $\mathcal{V}_g \times \{0\}$.

Second, distributional duality sends the attached scalar functional to a tensor density distribution with:

- unique bundle type;
- support;
- covariant transport;
- regularity order;
- divergence profile;
- finite additive and composite behavior.

Neither the attachment boundary nor these tensor consequences are present merely by writing a source label.

D Theorem-Use and Source Ledger

Source	Exact use	Prohibited promotion
AASC kernel [1]	Target, reference, standing, admissibility, irreversibility, no-repair order.	No metric, source tensor, or dynamics derived from the names alone.
RPI [2]	Upstream role architecture; fields, particles, deformation response, transport, vacuum, time, causality.	No local action or stress tensor imported automatically.
Mass/Inertia [3]	Deformation-response role and inertial restriction.	No gravitational mass, source tensor, or equivalence principle imported.
MGN1 [5]	Witness/quotient firewall; no numerical or coordinate selector.	No claim that central charge classifies full response.
Einstein dynamics [4]	Fixed metric source type and downstream source-ledger requirement.	No assumption that every upstream role already has a source functional.
Structural unification [14]	Joint-ledger typing and no projection backflow.	No inhabited joint continuation imported.

Source	Exact use	Prohibited promotion
Connection and covariant differentiation [6, 7]	Lawful metric transport and covariant derivative typing.	No source witness generated.
Distribution theory [8, 9]	Continuous-dual representation and support/order results.	No smoothness or physical source adequacy inferred.
Covariant Noether literature [10, 13]	Compatibility of action-based subclass and Ward identity.	No action primitivity or generic conserved global charge.
Distributional source literature [12]	Regularity/singularity comparator.	No claim that every distribution solves Einstein equations.

D.1 Dependency graph

$$\text{TargetAdeq} \longrightarrow \mathbf{K} \longrightarrow A^+ \longrightarrow \mathbf{I}_D \longrightarrow Q_{\text{resp}}.$$

$$Q_{\text{resp}}^{\text{src}} \subseteq Q_{\text{resp}}, \quad \widehat{Q}_{\text{src}} \twoheadrightarrow Q_{\text{resp}}^{\text{src}}, \quad \widehat{Q}_{\text{src}} \xrightarrow{q_g} \text{Src}_g^{\text{dist}} \xrightarrow{\text{div}} \mathcal{D}'_{\text{vec}}.$$

The attachment relation is formed from protocols already present in the complete upstream response signature. It determines the true base subtype before the structured source quotient and tensor representation are formed. The tensor and divergence objects do not feed backward.

E Claim-Status Matrix

Claim	Status	Boundary
Exact base subtype $Q_{\text{resp}}^{\text{src}}$	Proved by attachment theorem	Membership requires complete upstream metric incidence; arbitrary zero lift excluded.
Structured source quotient $\widehat{Q}_{\text{src}}$	Defined and audited	Retains attachment and witness data; not literally a subtype of Q_{resp} .
Operational-to-tensor representation	Proved	Distributional regularity by default.
Source quotient map q_g	Proved	Domain is the attachment-certified structured quotient only.
Naturality and support	Proved	Lawful metric redescrptions and fixed boundary class.
Ward/divergence equivalence	Proved	Compact interior probes or declared boundary pair.
Action/Hilbert compatibility	Proved conditionally	Action-presentable subclass only.
Composite total tensor	Proved	Finite completed certificate.
Internal exchange cancellation	Proved	Exact signed incidence balance.
Closed composite	Proved	Vanishing declared external residual.
Smooth Einstein source eligibility	Conditional	Smooth density and Einstein regularity certificates.

Claim	Status	Boundary
All RPI responses are source-bearing	Not claimed and formally false in the finite witness model	Source admission requires complete metric-incidence protocols.
Universal m_g/m_i	Not claimed	Later symmetry/charge module.
Upstream GR/QM continuation	Not claimed	Later continuation module.
Unique certificate orbit	Not claimed	Later fibre-exhaustion module.
PDE solution existence	Not claimed generally	Model-specific certificate required.
Weak equivalence	Not claimed	Existing TB branch remains separate.

F Standalone Hostile-Referee Checklist

A referee reviewing this module as an independent object should answer the following in order.

Dependency

1. Is the kernel/RPI/deformation-response order preserved?
2. Does any tensor or action retro-authorize the upstream response class?

Source-bearing type and attachment

1. Is the upstream metric-incidence relation independently contained in the response protocol signature rather than defined from the proposed witness?
2. Do totality, coherence, first-order vector compatibility, LF continuity, lawful transport, and metric-protocol completeness hold?
3. Does the attached functional have graph exactly equal to the upstream metric-response relation?
4. Is arbitrary zero attachment excluded while certified zero response remains admissible?
5. Is the true base subtype $Q_{\text{resp}}^{\text{src}} \subseteq Q_{\text{resp}}$ distinguished from the structured quotient $\widehat{Q}_{\text{src}}$?
6. Is the complete probe space mathematically well typed?

Representation

1. Is the continuous dual bundle correct?
2. Is the factor $1/2$ used consistently?
3. Are support, order, and regularity claims exact?
4. Is quotient descent representative independent?
5. Is naturality proved rather than asserted?

Ward closure

1. Is the distributional divergence sign correct?
2. Are boundary terms excluded or declared?
3. Is closed status distinguished from open/anomalous/off-shell status?

Composite closure

1. Does every source term occur exactly once?
2. Are cross terms explicit?
3. Are internal currents accompanied by signed incidence?
4. Does every incidence column sum to zero?
5. Is every nonzero external residual declared?
6. Does the software check only the finite claims it advertises?

Verdict format

The referee should issue:

- source-admission/representation verdict: correct / incorrect / unresolved;
- Ward-closure verdict: correct / incorrect / unresolved;
- composite-closure verdict: correct / incorrect / unresolved;
- software-correspondence verdict: correct / incorrect / unresolved;
- overall module verdict.

No capstone-level verdict should be inferred from this module review.

G Symbol Index

Symbol	Meaning
K_D	AASC kernel: admissibility, standing, reference, irreversibility.
I_D	Realized physical interior.
RespPres_D	Upstream admitted deformation-response presentations.
Q_{resp}	Upstream response quotient.
$\Sigma_{\text{resp}}(p)$	Complete upstream response protocol signature of presentation p .
$\tau_{g,p}$	Partial faithful metric-tangent map induced by the already licensed metric projection.
MetInc_p	Metric-incidence relation derived from upstream protocol incidence and $\tau_{g,p}$.
MetResp_p	Operational metric first-response relation of p .
$\mathfrak{a}_p$	Complete metric-attachment certificate for p .
$\text{Attach}_g(p, \mathfrak{a}, \mathcal{W})$	Relation asserting that $\mathcal{W}$ is the unique complete operational metric-response witness attached to p .
$\text{CompleteMetricWitness}_g(\mathfrak{w})$	Analytic support, covariance, Ward, regularity, and boundary certificate for an attached witness.
SrcPres_D	Structured source-bearing triples $(p, \mathfrak{a}_p, \mathfrak{w}_p)$.
$Q_{\text{resp}}^{\text{src}}$	True base subtype of response classes admitting at least one attached complete metric witness.
$\hat{Q}_{\text{src}}$	Structured source-witness quotient over $Q_{\text{resp}}^{\text{src}}$.
$\mathcal{V}_g$	Compactly supported smooth symmetric covariant metric variations.

Symbol	Meaning
$\mathcal{W}_p$	Attached operational metric first-response witness.
$\mathbf{T}_p$	Representing symmetric tensor-density distribution.
$\text{Src}_g^{\text{dist}}$	Distributional metric-source class.
Src_g^{sm}	Smooth metric-source subclass.
q_g	Natural quotient-faithful metric-source representation map with domain $\widehat{Q}_{\text{src}}$.
J_p	Declared Ward/exchange vector distribution.
$\mathcal{G}$	Finite source term registry of a composite.
A	Composite sector set.
E	Internal exchange channel set.
B_{ae}	Signed sector-channel incidence coefficient.
j_e	Internal exchange current distribution.
J_a^{ext}	External exchange current of sector a .
$\text{CompCert}(\mathcal{B})$	Completed composite source certificate.
$\simeq_{\text{law}}$	Lawful metric redescription/boundary equivalence.
$AASC - GRQM - MA - SRC - 1.0$	Frozen module endpoint identifier.

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